

★ WEEK 7 · PART 1 of 3

Cointegration: The Long-Run Anchor

Two Wanderers That Can't Drift Too Far Apart

The big idea · the equilibrium spread · the cure for spurious regression

Individually, two price series can wander without bound. Yet a special combination of them may stay put — a long-run equilibrium. That tethering is cointegration, and it anchors pairs trading.

1

PART 1 · COINTEGRATION

Tethered, Not Tied Down

The idea that each series wanders but their combination does not, the formal definition, the intuition, and how cointegration finally cures the spurious-regression trap of Week 2.

The Problem Granger Solved — Twice — Part 1

From exposing a trap to building its cure

In 1974, Clive Granger helped expose the spurious-regression trap (Week 2): unrelated trending series can look strongly related. Having revealed the danger, he then set out to fix it.

The trap he exposed

- Regressing one random walk on another gives a high R-squared.
- The relationship is a statistical illusion.
- It warned a generation off naive regressions of levels.

The question it raised

- If levels usually can't be regressed, are they ever safe to?
- Is there a case where a relationship in levels is REAL?
- Answering that led to cointegration.

The Problem Granger Solved — Twice — Part 2

Engle and Granger, 1987, and the lesson

Engle and Granger's 1987 paper defined cointegration and its error-correction counterpart — the exception where a levels relationship is genuine. It helped earn the 2003 Nobel.

What they showed

- Some $I(1)$ series share a genuine long-run equilibrium.
- A special combination of them IS stationary.
- Then regressing their levels is valid, not spurious.

The lesson for us

- Not all levels regressions are traps — cointegrated ones are real.
- The stationary combination is an economic anchor.
- That anchor is exactly what a pairs trade exploits.

From Short-Run Dynamics to a Long-Run Link

Where Week 6 left off

1

VAR needed stationary series

Week 6 modeled joint short-run dynamics — but a standard VAR requires differencing trending series first, which can discard their long-run relationship.

2

Some pairs share a long-run equilibrium

Two non-stationary prices may nonetheless move together forever, tied by economics. Differencing both would throw that tie away.

3

Cointegration models the tie directly

It captures the stable relationship in LEVELS, and error correction (Part 3) describes how deviations are pulled back. This is the pairs-trading engine.

Part 1 in one line: two series can each wander forever, yet their combination stays anchored.

The Big Idea of Cointegration

Each series wanders; their combination does not

Take two non-stationary series that each drift without bound. If some fixed linear combination of them is stationary, they are cointegrated — bound together by a long-run equilibrium.

Individually

- Each series is $I(1)$ — a wandering unit-root process.
- Neither has a mean it returns to.
- On its own, each looks unpredictable in level.

Together

- A specific combination cancels the shared wandering.
- What remains is stationary — it mean-reverts.
- That stationary combination is the equilibrium.

Cointegration, Formally

One line that captures the whole idea

$$X_t, Y_t \sim I(1) \quad \text{but} \quad z_t = Y_t - \beta X_t \sim I(0) \Rightarrow \text{cointegrated}$$

Reading it

- X and Y are each $I(1)$ — non-stationary.
- But the spread $z = Y - \beta X$ is $I(0)$ — stationary.
- That is the definition of cointegration.

The key word

- It only works for one special value of β .
- For any other β , the combination still wanders.
- Finding that β is the heart of the whole method.

The Integration Bookkeeping

Normally impossible — except when cointegrated

$$\underbrace{I(1)}_{Y_t} - \beta \underbrace{I(1)}_{X_t} = \underbrace{I(0)}_{\text{stationary spread}} \quad \text{only for the special } \beta$$

The usual rule

- Combining two $I(1)$ series generally gives another $I(1)$.
- The wandering does not cancel — it compounds.
- So a stationary combination should be impossible.

The exception

- Unless the two share the SAME stochastic trend.
- Then the right β subtracts one trend from the other.
- The trends cancel, leaving a stationary remainder.

The Drunk and Her Dog

The metaphor that makes cointegration click

A classic image: a drunk walks home randomly; her dog wanders randomly too. Each path is a random walk. But a leash keeps them close — so the DISTANCE between them is stationary.

The two random walks

- The drunk's position wanders unpredictably.
- The dog's position wanders unpredictably too.
- Neither one, alone, returns to any fixed spot.

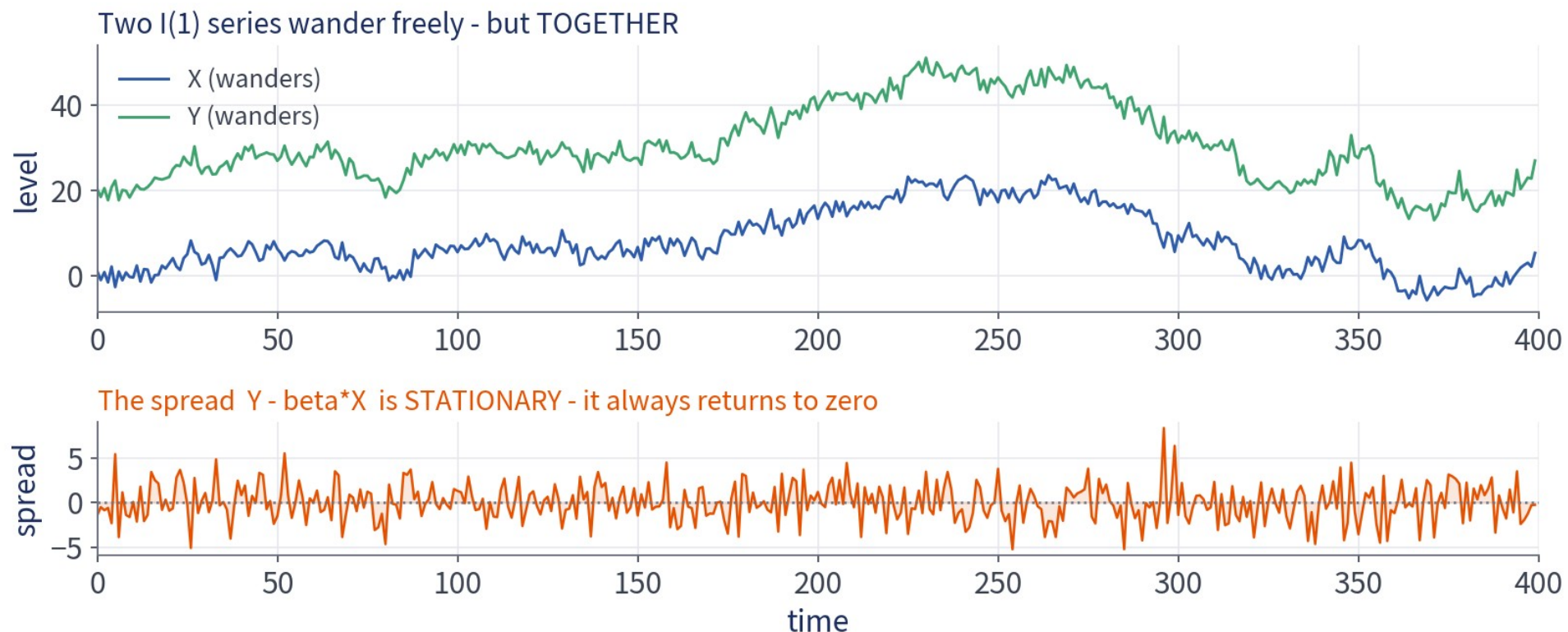
The leash = cointegration

- The leash ties them: they cannot drift far apart.
- Their separation mean-reverts around the leash length.
- That bounded gap is the stationary spread.

Tethered, Not Tied Down

Both series wander freely, yet always together (top). Their spread $Y - \beta X$ is stationary — it repeatedly returns to zero (bottom). That returning spread is the anchor.

Tethered, not tied down: cointegration



The Equilibrium Spread

The stationary combination, and what it means

What the spread is

- The stationary combination $z = Y - \beta X$.
- It measures the deviation from the long-run equilibrium.
- Zero means the pair is 'fairly' priced relative to each other.

Why it is powerful

- A positive spread $\rightarrow$ Y is rich, X is cheap (relatively).
- A negative spread $\rightarrow$ the reverse.
- Because it mean-reverts, extremes are trading opportunities.

The Spurious-Regression Cure

Week 2's villain, finally defeated

Spurious regression warned us never to regress one trending series on another. Cointegration is the precise exception: when the residual is stationary, the regression is genuine.

The spurious case (Week 2)

- Two unrelated random walks give a high but meaningless R-squared.
- The regression residual itself wanders — it is $I(1)$.
- The relationship is a mirage.

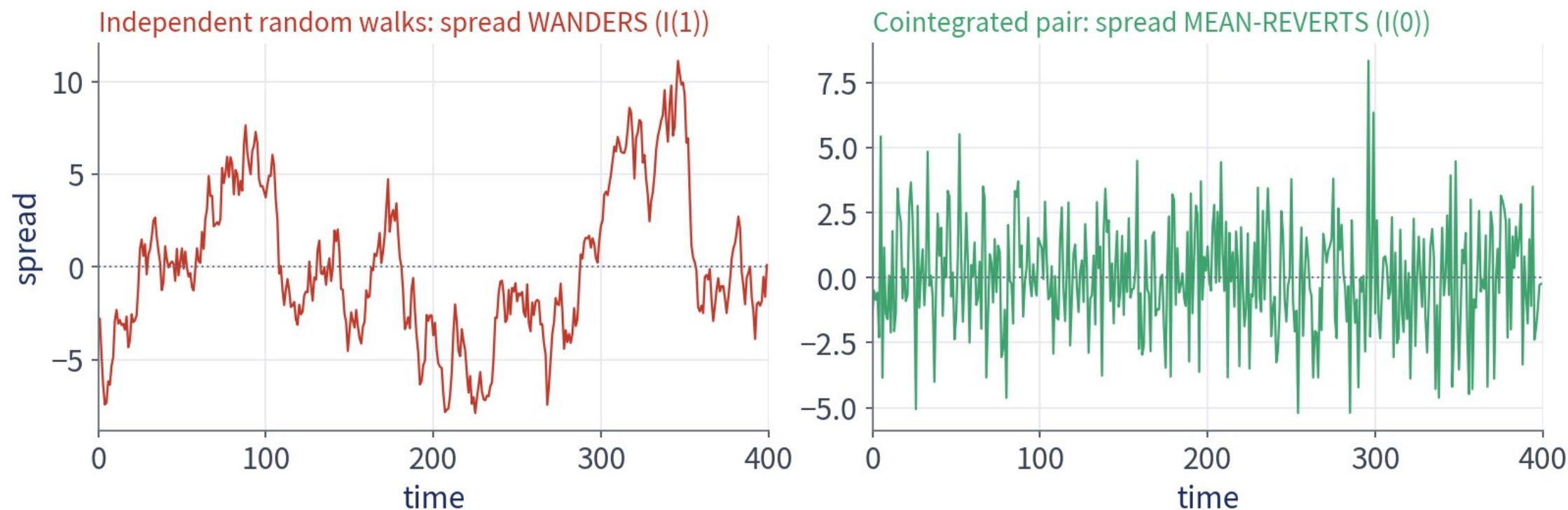
The cointegrated case

- The residual is stationary — $I(0)$.
- That stationarity is the certificate of a real relationship.
- So testing the residual is exactly how we tell them apart (Part 2).

Spurious vs Genuine

The residual is the tell. Independent random walks leave a spread that keeps wandering (left, $I(1)$); a truly cointegrated pair leaves a spread that mean-reverts (right, $I(0)$).

The tell: a genuine equilibrium spread is stationary; a spurious one is not



Cointegration vs Correlation

Related ideas, but not the same

Correlation

- Measures short-run co-movement of returns.
- Can be high one month and gone the next.
- Says nothing about a long-run tie in levels.

Cointegration

- A long-run equilibrium relationship in LEVELS.
- Deviations are temporary and get corrected.
- Two series can be cointegrated yet weakly correlated day-to-day.

The Cointegrating Vector

β is the hedge ratio in disguise

cointegrating vector $(1, -\beta)$: $z_t = Y_t - \beta X_t$ is the equilibrium error

What β means

- It is the weight that makes the combination stationary.
- The pair $(1, -\beta)$ is the cointegrating vector.
- z is the equilibrium error — the spread we watch.

The trading reading

- β is how many units of X hedge one unit of Y .
- It defines the market-neutral spread position.
- Estimate it once; it anchors the whole strategy.

Why Finance Loves Cointegration

A rare source of genuine mean reversion

Prices are close to random walks — hard to forecast. But a cointegrated SPREAD mean-reverts, giving a statistically grounded signal that survives where single-asset forecasting fails.

What it provides

- A mean-reverting quantity built from non-mean-reverting prices.
- A market-neutral position: long one, short the other.
- An economic anchor, not just a fitted pattern.

Where it shows up

- Relative-value and statistical-arbitrage strategies.
- Index-arbitrage and futures-basis trades.
- The whole discipline of pairs trading.

Examples in Markets

Where cointegration plausibly holds

Strong candidates

- Spot and futures on the same underlying.
- Dual-listed shares of the same company.
- Two firms in the same industry with shared drivers.

The economic logic

- An arbitrage or substitution force ties them together.
- When they diverge, that force pulls them back.
- No such force → no cointegration, however correlated.

The Common-Trend View

One shared random walk underneath

Cointegrated series can be seen as one shared stochastic trend plus stationary noise. The right β removes the common trend, and only the stationary parts remain — the spread.

The decomposition

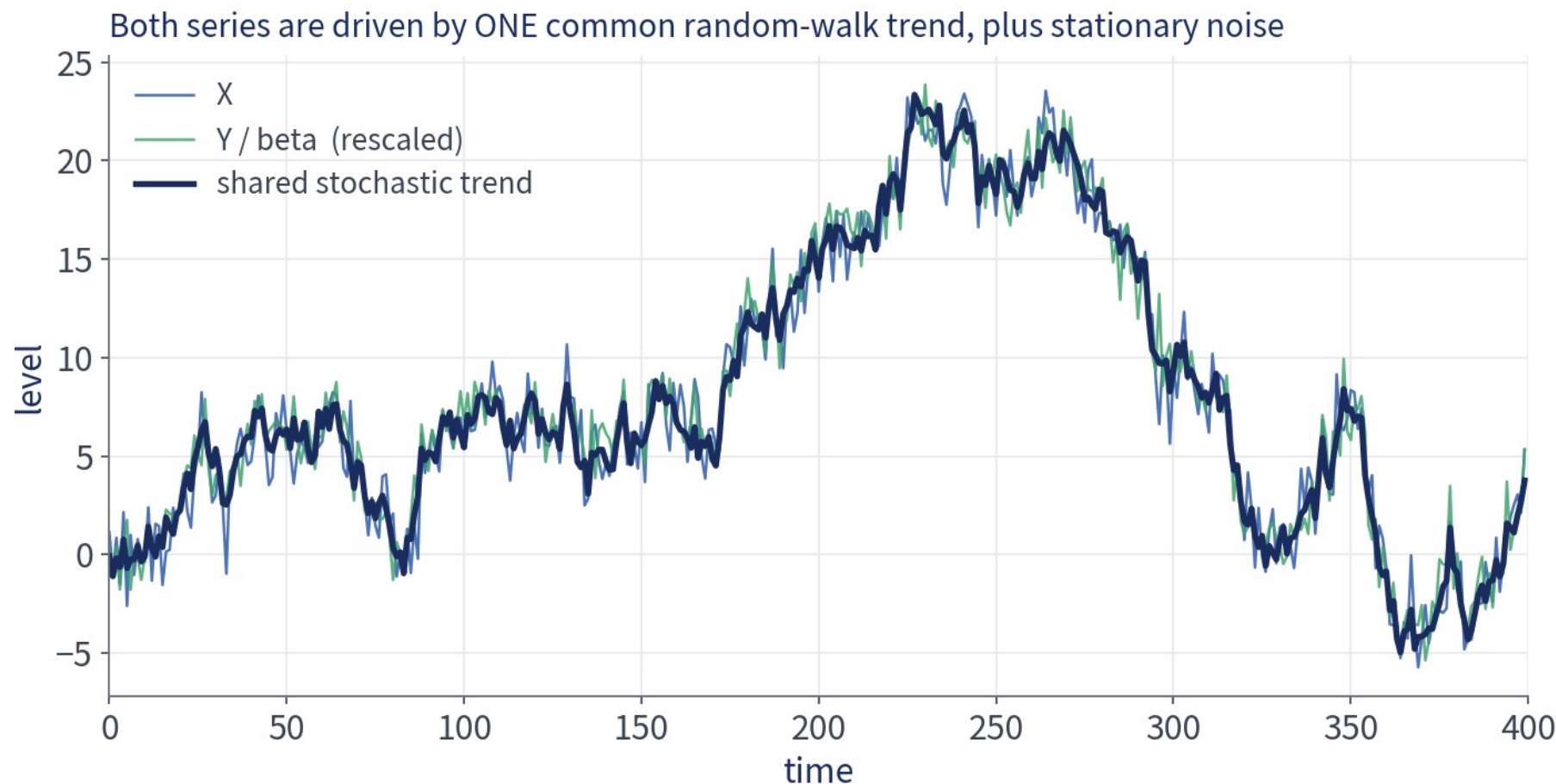
- Each series = a common random-walk trend + its own noise.
- The common trend is the source of the non-stationarity.
- It is SHARED, so it can be cancelled.

Why this clarifies it

- Cointegration = 'fewer trends than series'.
- Two series driven by one trend are cointegrated.
- Subtracting the shared trend leaves the stationary spread.

One Common Trend Underneath

Both series track a single shared random-walk trend (dark line), each adding only stationary noise. That shared trend is what the cointegrating combination cancels.



When Two Series Are NOT Cointegrated

No shared trend, no anchor

The signs

- Every linear combination still wanders — the spread is $I(1)$.
- There is no economic force binding them.
- High correlation can still occur, but there is no equilibrium.

Why it matters

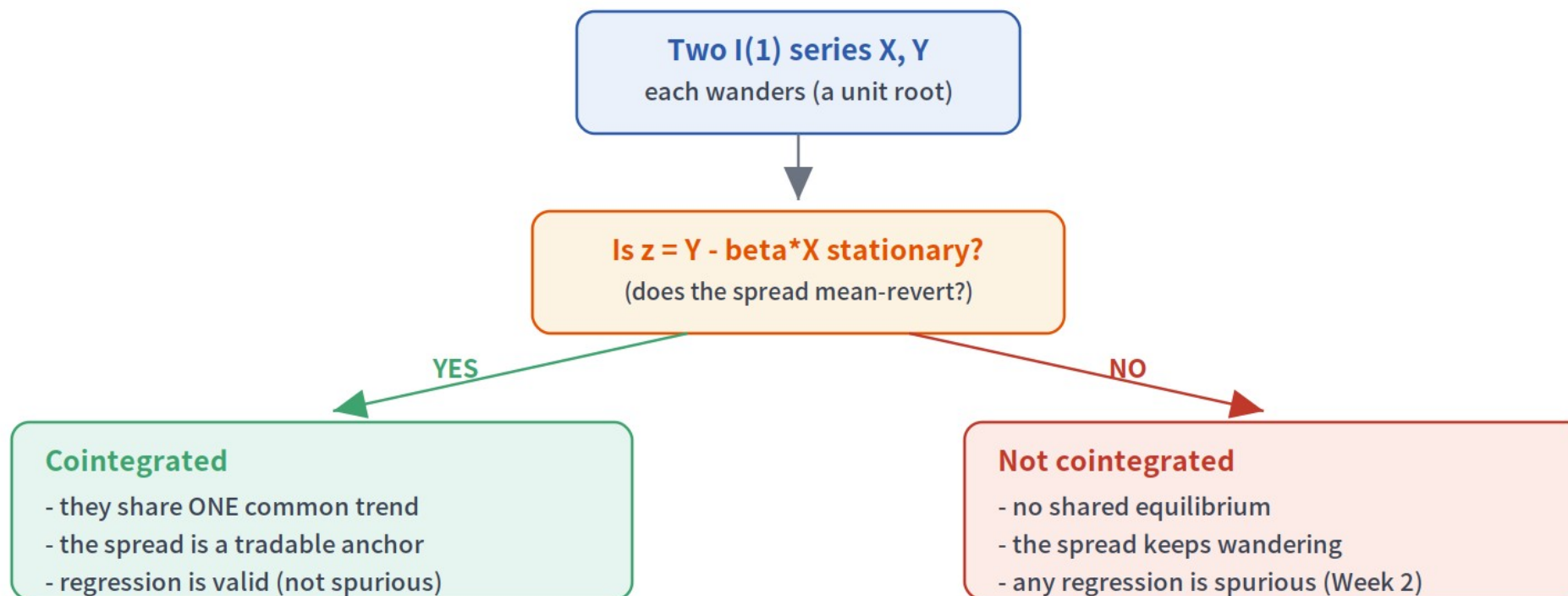
- Trading a non-cointegrated 'spread' is trading a random walk.
- It can drift away and never come back.
- Testing (Part 2) is what protects you from this.

Is a Pair Cointegrated?

Two $I(1)$ series are cointegrated only if some combination $Y - \beta X$ is stationary. If yes, they share a trend and give a tradable anchor; if no, any regression is spurious.

Is a Pair Cointegrated?

Two wandering series are cointegrated only if some linear combination of them stops wandering.



From Concept to Test to Trade

How the rest of the week unfolds

Part 2 — testing

- Engle-Granger: regress, then test the residual for stationarity.
- Johansen: a system test for several series at once.
- Read the cointegration rank — how many equilibria hold.

Part 3 — trading

- Error correction: the force that restores the equilibrium.
- The VECM and the speed of adjustment.
- Pairs trading Stage II, built on the cointegration spread.

Visualize a Candidate Pair

Direct Claude Code to see the anchor before testing

GOAL Take a plausible pair, estimate the spread, and see whether it looks stationary — the visual precursor to a formal test.

[INPUT] Prompt · Claude Code

Write and run `coint_intro.py` that:

1. downloads two related price series (e.g. two ETFs in the same sector, or a stock and its index) and confirms each is $I(1)$;
 2. regresses one on the other to estimate β , and forms the spread $z = Y - \beta X$;
 3. plots both prices together and the spread below, marking its mean;
 4. eyeballs whether the spread mean-reverts.
- Tell me: does this pair look cointegrated, and what is the estimated β (hedge ratio)?

How to work with it

- Both series should each be non-stationary ($I(1)$).
- A spread hugging its mean suggests cointegration.
- A spread that drifts away suggests none.
- Keep it; Part 2 turns eyeballing into a test.

Deliverable: `coint_intro.py` + a prices-and-spread plot with the estimated β .

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1	Two $I(1)$ prices have a stationary spread $Y - \beta X$. What are they?	Cointegrated
Q2	How does cointegration escape the spurious-regression trap?	Stationary residual
Q3	Can two series be highly correlated but not cointegrated?	Yes — different ideas

All three come from the definition, the stationary spread, and the correlation contrast.

Three Misconceptions to Avoid

Where cointegration intuition slips

Common myths

- "High correlation means cointegration."
- "Any two trending series can be paired."
- "A cointegrated spread can never drift far."

The reality

- Correlation is short-run; cointegration is a long-run levels tie.
- Only pairs sharing a common trend are cointegrated — most aren't.
- Spreads mean-revert on average, but can stray far first (Part 3).

Three Things to Carry Forward

Part 1 in one screen

1 Cointegration = a stationary combination of wanderers

Two $I(1)$ series are cointegrated if some combination $Y - \beta X$ is stationary. They share one common trend, and the spread is their long-run equilibrium error.

2 It cures spurious regression

A levels regression is genuine precisely when its residual is stationary. That stationary spread is the certificate — and testing it is how we confirm cointegration.

3 The spread is a tradable anchor

Prices barely mean-revert, but a cointegrated spread does. That is a rare, economically grounded signal — the foundation of pairs trading.

Next — Part 2: testing for cointegration with Engle-Granger and the Johansen system approach.

★ WEEK 7 · PART 1 · PRACTICE

Cointegration, Worked by Hand

Find the Hedge Ratio, Then Tell It From Mere Correlation

The equilibrium spread and the cointegration test

Part 1's core idea, with the arithmetic shown. We estimate the hedge ratio that links two wandering prices into a steady spread, then see why two highly correlated series can still fail to cointegrate - the distinction that keeps a pairs trade from blowing up.

1P

PART 1 · APPLICATIONS

One Steady Spread From Two Drifters

Case A: estimate the cointegrating vector - the hedge ratio - and form the equilibrium spread, showing it stays flat while both prices climb. Case B: cointegration versus correlation, and why only one of two correlated pairs is tradable. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Each series wanders, the combination doesn't

Two prices can each be non-stationary, yet a particular linear combination of them is stationary - that's cointegration.

2

The cointegrating vector is a hedge ratio

The coefficient beta that makes the spread stationary is exactly how much of one asset to hold against the other.

3

Cointegration is not correlation

Two series can move together in the short run yet drift apart forever. Only a stationary spread supports a pairs trade.

Everything here is a calculator exercise: one regression, one spread, two comparisons.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from two price series.
- Estimate the spread, or compare two pairs.
- Then read whether it reverts.

What you should watch

- How beta turns two prices into one spread.
- How the spread stays bounded while prices climb.
- How correlation can mislead you.

The Hedge Ratio and the Spread

Two wandering prices, one steady link

Two related stocks each drift like a random walk, yet they stay tethered. Regressing one price on the other gives beta - the cointegrating vector - and the residual is the spread. We estimate beta by hand, form the spread, and confirm it hovers in a narrow band even as both prices climb ten points

The data

- Stock X drifts from 100 up to 112.
- Stock Y roughly tracks about twice X.
- Eight observations.

What we'll do by hand

- Regress Y on X for beta.
- Form the spread Y minus beta-X.
- Check that it stays bounded.

The Cointegrating Relationship

A stationary spread links two drifters

$$\underbrace{Y_t}_{I(1)} = \alpha + \beta \underbrace{X_t}_{I(1)} + \underbrace{s_t}_{\text{stationary spread}} \quad (\text{two wandering prices, one steady combination})$$

The pieces

- Both prices are individually non-stationary.
- Beta scales one to match the other.
- What's left over is the spread.

The magic

- The spread is stationary - it mean-reverts.
- So the two prices never wander far apart.
- Beta is the tether.

Estimate the Hedge Ratio

Regress one price on the other

$$\beta = \frac{\text{Cov}(Y,X)}{\text{Var}(X)} = \frac{33.23}{16.98} = \mathbf{1.96}, \quad \alpha = \bar{Y} - \beta\bar{X} = \mathbf{5.44}$$

The calculation

- Covariance of Y and X is 33.23.
- Variance of X is 16.98.
- Their ratio is beta = 1.96.

The meaning

- Hold one share of Y against 1.96 of X.
- That's the hedge ratio.
- The intercept sets the spread's level.

Form the Spread

Bounded while the prices climb

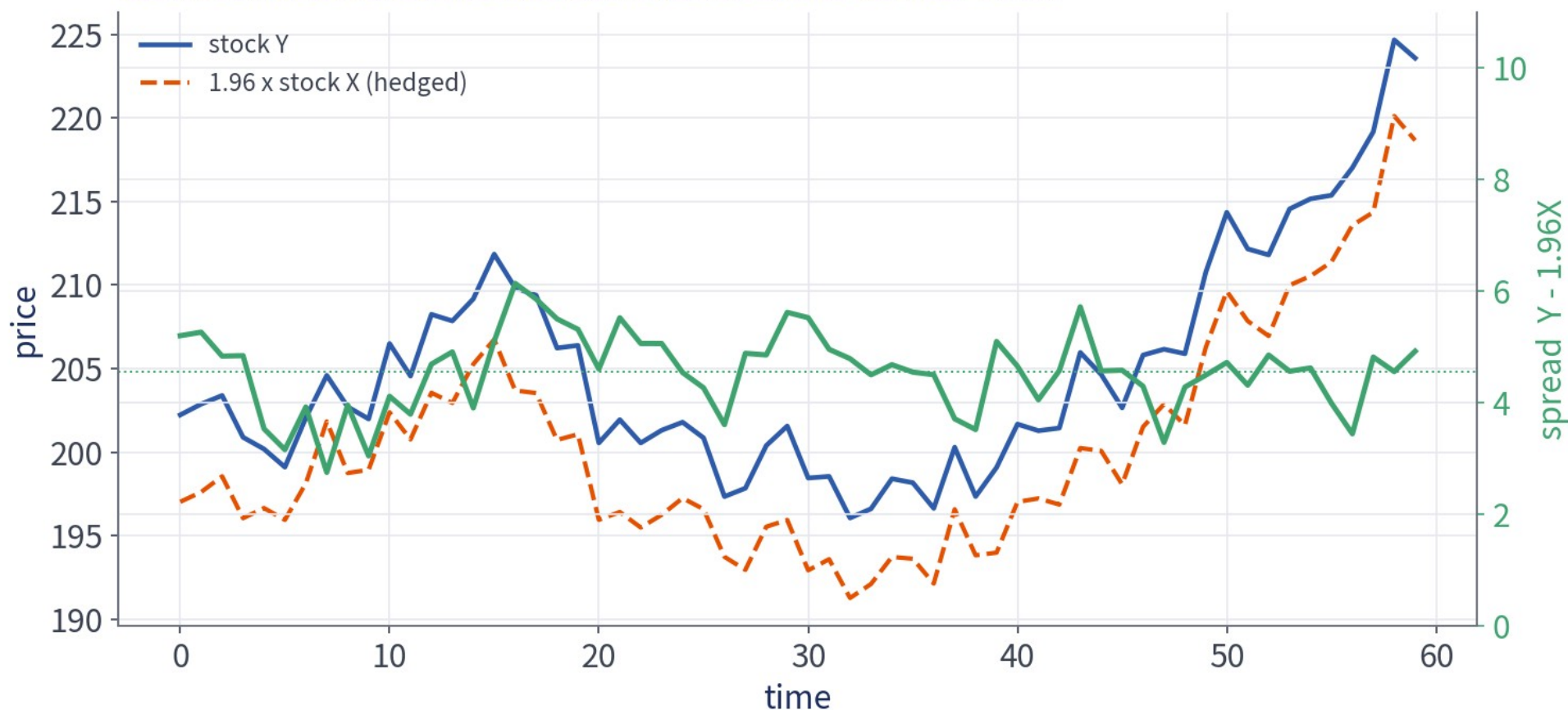
X	100	103	107	110	112
Y	201	208	215	221	224
spread $Y - 1.96X$	5.0	6.1	5.3	5.4	4.5

As X climbs from 100 to 112, the spread never leaves the 4.1-to-6.1 band, hovering near 5.2. Both prices trend, but their hedged combination is flat - the signature of cointegration.

Prices Climb, Spread Stays Flat

Stock Y and its hedged partner 1.96-times-X climb together over time, wandering up the chart. But the spread between them (green, right axis) barely moves - it mean-reverts inside a tight band. That flat green line is the equilibrium the two prices are tethered to

Case A: both prices wander and climb, but the spread stays flat and mean-reverts



Beta Is the Trade

From regression to hedge

What beta gives you

- The exact hedge ratio for a pairs trade.
- The spread you actually trade around.
- An equilibrium to bet on reverting to.

Why it matters

- When the spread stretches, you fade it.
- When it reverts, you profit.
- The whole trade rests on that stationarity.

Cointegration vs Correlation

Moving together is not enough

Two pairs of stocks both show a correlation near 0.997 - textbook lockstep. Yet only one is cointegrated. In the first, the spread reverts inside a bounded range; in the second, it drifts steadily wider and never comes back. Correlation looks the same for both; only the spread's behavior reveals which pair is tradable

The two pairs

- Both partners trend up strongly.
- Both correlate about 0.997.
- Indistinguishable on correlation alone.

What we'll do by hand

- Track each pair's spread over time.
- See one revert, the other drift.
- Read which is tradable.

The Crucial Distinction

Correlation is short-run, cointegration is long-run

correlation ≈ 0.997 for BOTH pairs, but only one has a stationary spread (cointegration $\neq$ correlation)

Correlation

- Measures short-run co-movement.
- Can be high even without a tether.
- Says nothing about drifting apart.

Cointegration

- Requires a stationary spread.
- Guarantees the gap stays bounded.
- That's what a pairs trade needs.

The Cointegrated Pair

A spread that reverts

time	1	2	3	4	5	6	7	
spread	+0.5	-0.8	+0.3	-0.6	+0.7	-0.4	+0.5	-0

The spread crosses zero again and again, staying inside a narrow band from -0.8 to +0.7. It reverts - a stationary spread - so the pair is tradable: fade the extremes, collect on the return.

What you see

- The spread flips sign every step.
- It never leaves the -0.8 to +0.7 band.
- A clear, bounded mean-reversion.

Why it is tradable

- Fade the spread when it stretches wide.
- Collect as it snaps back to zero.
- Stationary spread, repeatable edge.

The Correlated-Only Pair

A spread that drifts

time	1	2	3	4	5	6	7	
spread	0.0	0.6	1.3	2.1	3.0	4.2	5.5	6.9

Here the spread only grows - 0 to 6.9 - never turning back. Despite a 0.997 correlation, the gap wanders without bound: not cointegrated. A pairs trade here would keep losing as the spread widens.

What you see

- The spread climbs every single step.
- It runs from 0 all the way to 6.9.
- No sign of turning back.

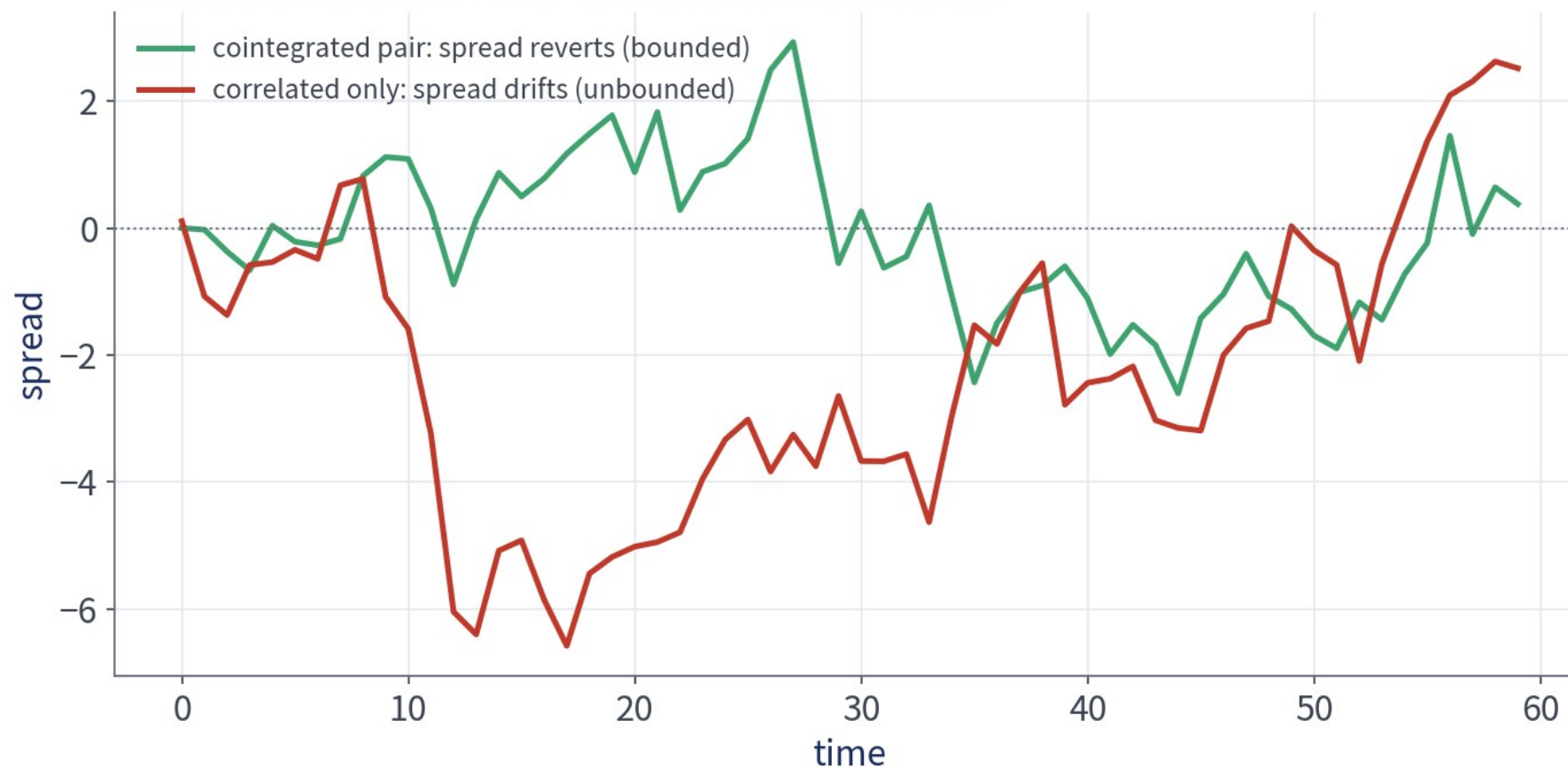
Why it is not tradable

- A short spread trade keeps losing.
- The gap widens without a mean to revert to.
- High correlation gave no warning.

One Reverts, One Runs Away

Both pairs correlate near 0.997, yet their spreads could not differ more. The green spread oscillates around zero - stationary, tradable. The red spread drifts steadily away and never returns - a random walk in disguise. Correlation could not tell them apart; the spread does.

Case B: both pairs correlate ~ 0.997 - but only the green spread reverts



Why the Distinction Saves You

Correlation is a trap here

The trap

- Pairing two stocks just because they correlate.
- Assuming the spread will always come back.
- Watching it drift and holding on.

The discipline

- Trade only cointegrated pairs.
- Test the spread for stationarity first.
- Correlation is necessary, not sufficient.

One Shared Trend Underneath

The anchor that makes it work

Cointegration means two prices share a single underlying random walk - a common trend. Beta scales one series so that shared trend cancels, leaving only a stationary spread. That's why the hedged combination stays flat, and why a genuine pairs trade rests on cointegration, not correlation: the shared trend is the anchor, and

$$Y_t = \gamma T_t + u_t, \quad X_t = T_t + v_t \Rightarrow Y_t - \gamma X_t = u_t - \gamma v_t \text{ (shared trend } T_t \text{ cancels)}$$

Reproduce the Spread and the Test

Direct Claude Code to check your arithmetic

GOAL Estimate the hedge ratio and spread, then distinguish cointegration from correlation.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - hedge ratio:

- * regress $Y=[201,204,211,208,215,221,217,224]$ on $X=[100,102,105,103,107,110,108,112]$;
- * print beta, alpha, and the spread $Y - \text{beta} * X$.

CASE B - cointegration vs correlation:

- * build a reverting spread and a drifting spread;
- * print the correlation of each pair (both ~ 0.997);
- * run an ADF test on each spread.

Then plot the prices with the spread, and the two spreads side by side.

Check against the slides

- beta 1.96, alpha 5.44.
- spread in $[4.1, 6.1]$.
- both correlations ~ 0.997 .
- only the reverting spread is $I(0)$.

Deliverable: `part1_practice.py` whose printed numbers match the spread and the two pairs.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the cointegrating vector beta represent?

**The hedge ratio
between the two
assets**

Q2 How do you know a pair is cointegrated?

**Their spread is
stationary - it mean-
reverts**

Q3 Why isn't high correlation enough?

**Correlated series can
still drift apart
forever**

All three come from the hedge-ratio regression and the cointegration-versus-correlation test.

Three Pitfalls to Avoid

Where cointegration trips people

Common mistakes

- "High correlation means a tradable pair."
- "Any linear combination gives the spread."
- "The spread will always come back."

The reality

- Correlation can hide a spread that drifts forever.
- Only the cointegrating beta makes it stationary.
- Test for stationarity - don't assume reversion.

Two Calculations to Remember

Part 1 applications in one screen

1

The hedge ratio makes the spread

Regressing Y on X gave $\beta = 1.96$ - the cointegrating vector. The spread $Y - 1.96X$ stayed in a tight 4.1-to-6.1 band while X climbed from 100 to 112. Beta is the hedge ratio, and the flat spread is the equilibrium you trade around.

2

Correlation is not cointegration

Two pairs both correlated near 0.997 behaved oppositely: one spread reverted inside a bounded range, the other drifted from 0 to 6.9 and never returned. Only a stationary spread supports a pairs trade - correlation alone would have misled you.

3

A shared trend is the anchor

Cointegration means the two prices ride one common random walk, which beta cancels to leave a stationary spread. That shared trend is what keeps the pair tethered - and what a genuine trade depends on.

Next — Part 2 practice: formally testing for cointegration with Engle-Granger and Johansen.

★ WEEK 7 · PART 2 of 3

Testing for Cointegration

From Eyeballing the Spread to a Formal Verdict

Engle-Granger two-step · the Johansen system test · the cointegration rank

A spread that looks stationary might just be lucky. Part 2 makes the judgment rigorous: two formal tests that decide whether a genuine long-run equilibrium really exists.

2

PART 2 · TESTING

Is It Really Cointegrated?

The Engle-Granger two-step residual test, the Johansen system approach and its cointegration rank, when to use each, and the practical pitfalls of testing.

From a Picture to a Test

Part 1 gave the idea; Part 2 makes it rigorous

1

We can now recognize cointegration

Two $I(1)$ series whose combination is stationary share a long-run equilibrium. The spread is our anchor — but a plot alone is not proof.

2

Eyeballing is not enough

A spread can look mean-reverting by chance over a short sample. We need a formal test with proper critical values.

3

Two standard tests do the job

Engle-Granger for a pair, Johansen for a system. Both ask the same question — is there a stationary combination? — with statistical rigor.

Part 2 in one line: prove the equilibrium is real before you trade it.

Prerequisite: Confirm Each Series Is $I(1)$

Cointegration is a statement about $I(1)$ series

Before testing for cointegration, verify that each series is individually $I(1)$ — non-stationary in levels, stationary in differences. Cointegration only makes sense for such series.

The check

- Run a unit-root test (ADF/KPSS) on each level — expect non-stationary.
- Run it on each difference — expect stationary.
- Both series should be $I(1)$ before you proceed.

Why it matters

- If a series is already stationary, cointegration is the wrong tool.
- If a series is $I(2)$, the setup changes entirely.
- Getting the integration order right is step zero.

The Testing Landscape

Two classic ways to test for cointegration

Engle-Granger

- A two-step, single-equation procedure.
- Regress one series on the other, then test the residual.
- Simple and intuitive — ideal for a single pair.

Johansen

- A system approach based on a vector error-correction model.
- Handles many series and finds how MANY equilibria exist.
- More powerful and symmetric, but more involved.

Step 1: Estimate the Long-Run Relationship

A simple regression of one level on the other

Step 1: $Y_t = \alpha + \beta X_t + u_t$ (OLS estimate of the long-run relationship)

What you do

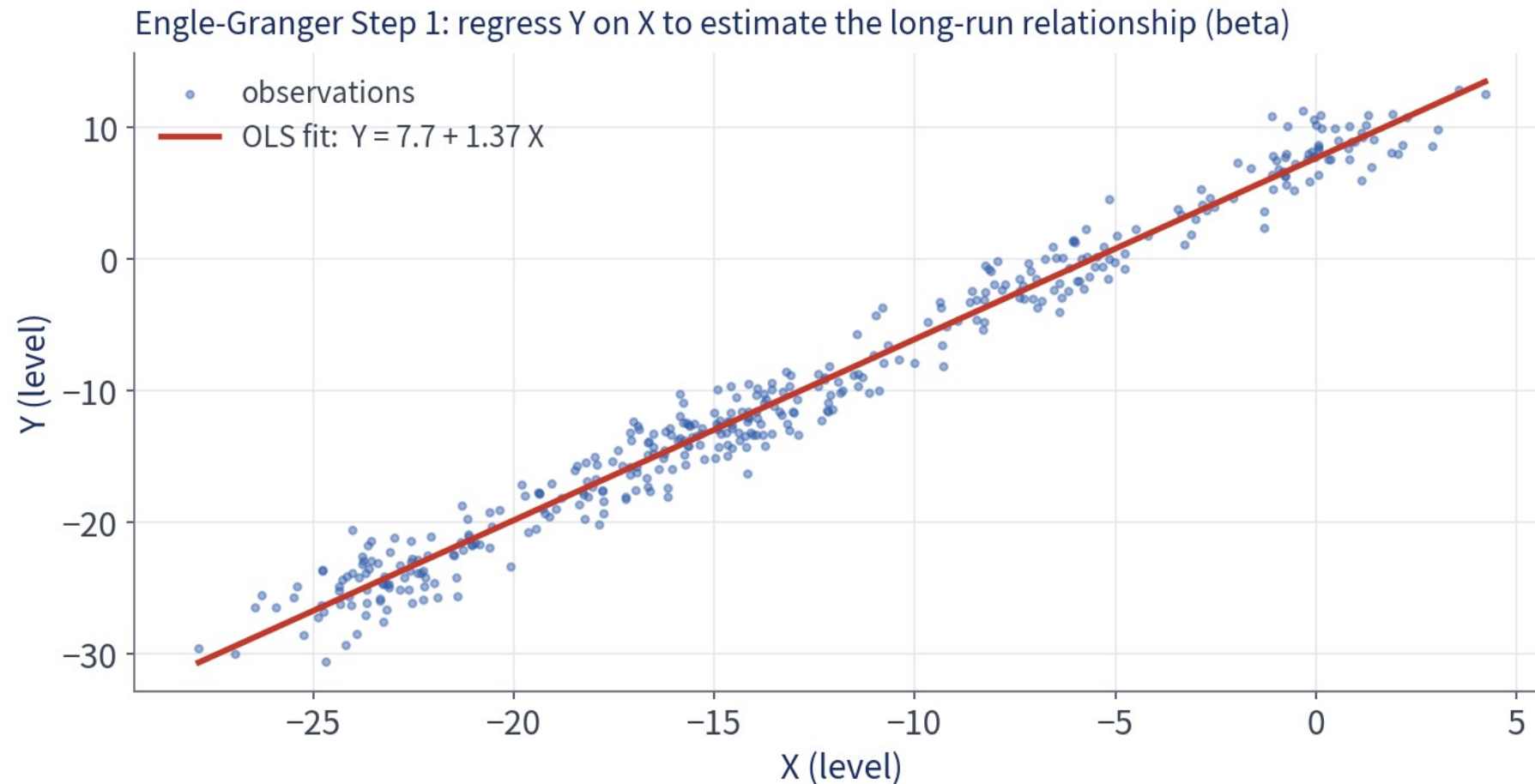
- Regress Y on X by ordinary least squares.
- The slope estimates the cointegrating coefficient β .
- The residual is the estimated equilibrium error.

What it gives

- A candidate long-run relationship between the levels.
- The spread $z = Y - \beta X$ to be tested next.
- This β is also the hedge ratio for trading (Part 3).

Step 1, Visualized

The regression line through the two levels estimates the long-run relationship. Its slope is β — the weight that (if the pair is cointegrated) makes the spread stationary.



Step 2: Test the Residual for Stationarity

The residual is the verdict

Step 2: ADF test on $\hat{u}_t$: H_0 : $\hat{u}_t$ has a unit root (NOT cointegrated)

What you do

- Run a unit-root (ADF) test on the step-1 residual.
- The null is that the residual has a unit root — NOT cointegrated.
- Reject the null $\rightarrow$ the residual is stationary $\rightarrow$ cointegrated.

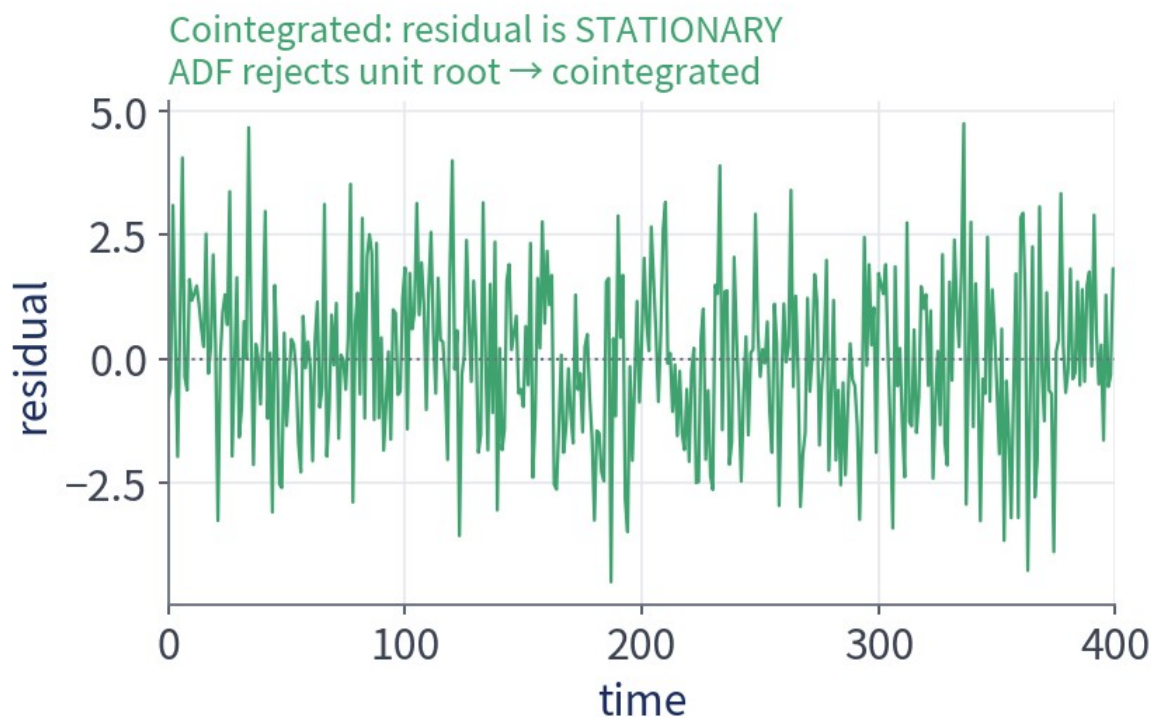
Why this is the whole test

- A stationary residual is the certificate of a real relationship.
- A wandering residual means the regression was spurious.
- Everything hinges on this one stationarity test.

Step 2, Visualized

The residual decides. For a cointegrated pair it mean-reverts and ADF rejects a unit root (left); for an unrelated pair it wanders and ADF cannot reject (right).

Engle-Granger Step 2: run an ADF test on the residual



Why Special Critical Values

The residual is estimated, not observed

The step-2 residual comes from a fitted regression, so ordinary ADF critical values don't apply. Engle-Granger uses its own, more demanding critical values.

The issue

- OLS chooses β to make the residual as small as possible.
- That biases it toward looking stationary.
- Standard ADF thresholds would over-reject the null.

The fix

- Use Engle-Granger (MacKinnon) critical values instead.
- They are stricter, correcting for the estimated β .
- Software applies the right ones automatically — just don't use plain ADF.

Which Series Is the Dependent One?

Y on X, or X on Y?

The problem

- Regressing Y on X and X on Y give different β estimates.
- In finite samples the test result can differ too.
- The method is not symmetric in the two variables.

How to handle it

- Try both orderings and check the conclusions agree.
- Prefer the more 'exogenous' series as the regressor if there is one.
- Or sidestep the issue entirely with Johansen.

The Limits of Engle-Granger

Great for a pair, awkward beyond it

Engle-Granger is simple and intuitive, but it can find only ONE relationship, works cleanly for just two series, and depends on which variable you place on the left.

What it can't do

- It finds at most one cointegrating relationship.
- With three or more series there may be several.
- It cannot tell you HOW MANY equilibria exist.

The remedy

- A system test that treats all series symmetrically.
- One that reports the NUMBER of relationships.
- That is exactly the Johansen approach.

The Johansen Test

A system approach for many series

Johansen fits a vector error-correction model directly and asks how many independent stationary combinations exist among the series — treating them all symmetrically.

What it does

- Works from the VECM, not a single regression.
- Estimates the number of cointegrating relations at once.
- Returns each cointegrating vector, not just one.

Why it is preferred for systems

- No arbitrary choice of dependent variable.
- It handles three or more series naturally.
- It is generally more powerful than Engle-Granger.

Where Johansen Looks

The rank of one matrix tells the whole story

$$\Delta \mathbf{Y}_t = \Pi \mathbf{Y}_{t-1} + \sum_i \Gamma_i \Delta \mathbf{Y}_{t-i} + \boldsymbol{\epsilon}_t, \quad \text{rank}(\Pi) = r$$

The key object

- Write the system as changes explained by past levels.
- The matrix on the lagged levels carries the equilibrium info.
- Its RANK equals the number of cointegrating relations.

The intuition

- If that matrix is zero, nothing ties the levels — no cointegration.
- If it has full rank, the series were stationary all along.
- In between: the rank counts the equilibria (Part 3 uses this).

The Cointegration Rank r

How many equilibria hold the system together?

$r = 0$: no cointegration $0 < r < k$: r relations $r = k$: all stationary

Reading r

- $r = 0$: no cointegration — nothing ties the series.
- $0 < r < k$: exactly r long-run relationships.
- $r = k$: the series were all stationary to begin with.

Why the count matters

- It tells you how many tradable equilibria exist.
- With k series you can have up to $k - 1$ relationships.
- Johansen finds this number; Engle-Granger cannot.

Determining the Rank

Step down until you can't reject

The procedure

- Test the null $r = 0$; if rejected, there is at least one relation.
- Then test $r \leq 1$, $r \leq 2$, and so on.
- Stop at the first null you cannot reject — that is the rank.

What you compare

- A test statistic against a critical value at each step.
- A statistic above the critical value rejects that null.
- The chain of rejections pins down r .

Trace vs Maximum-Eigenvalue

Two flavors of the Johansen test

The trace test

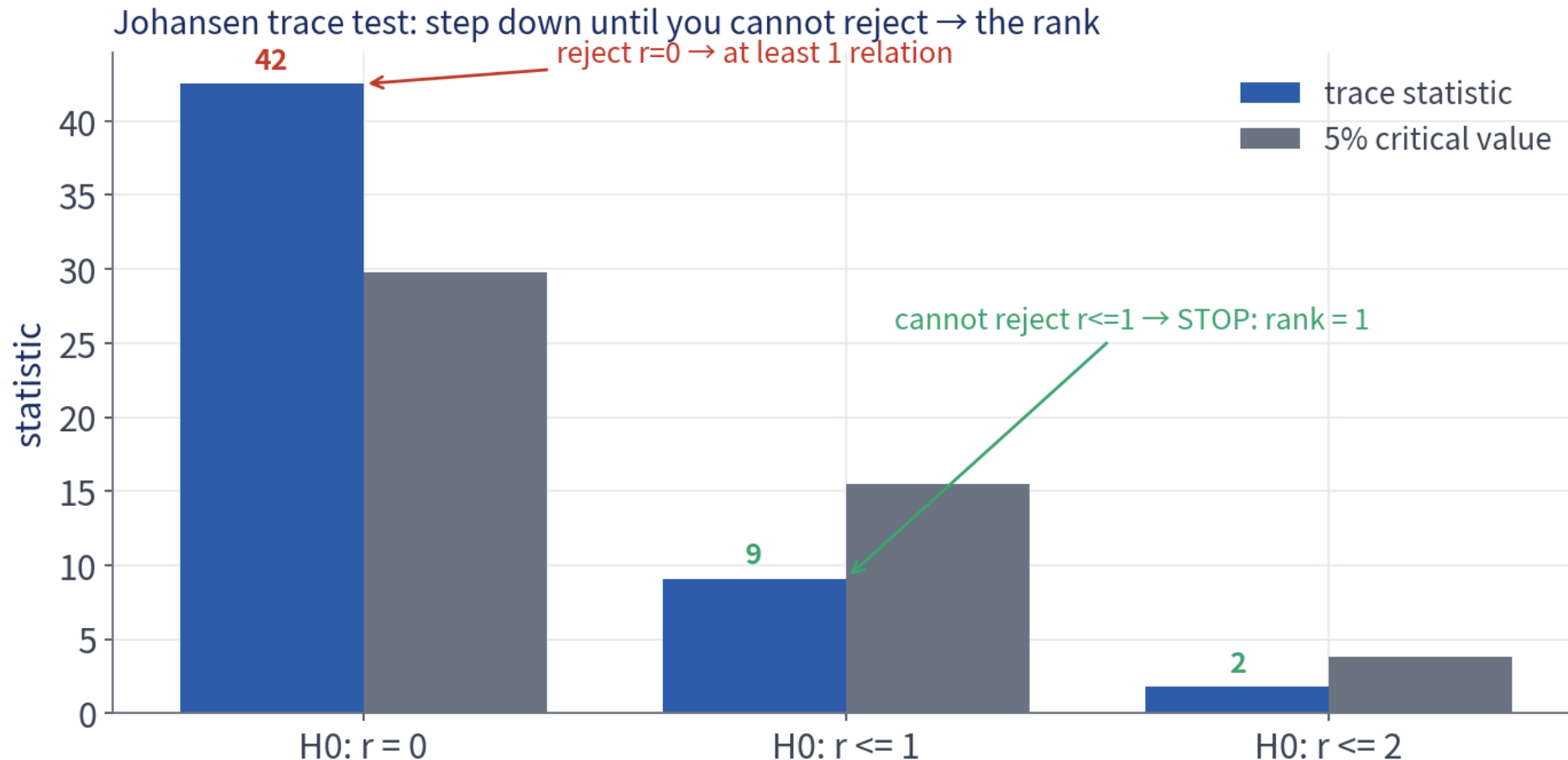
- Tests 'at most r relations' against 'more than r '.
- It pools information across the remaining directions.
- The most commonly reported version.

The max-eigenvalue test

- Tests 'exactly r ' against 'exactly $r + 1$ '.
- More focused on the single next relationship.
- The two usually agree; when they don't, investigate.

Reading the Trace Test

Reject $r = 0$ (statistic above the critical value) — so at least one relation exists. Fail to reject $r \leq 1$ — so stop: the cointegration rank is one.



Engle-Granger vs Johansen

Engle-Granger is a quick two-step check for a single pair; Johansen is a symmetric system test that finds how many equilibria exist. Match the test to the problem.

Engle-Granger vs Johansen

Pick the test that matches the number of series - and what you need from it.

Engle-Granger

a two-step, single-equation test

Best for

- exactly TWO series (one pair)
- a quick, intuitive check

How it works

1. regress Y on X (get beta)
2. ADF-test the residual

Limits

- finds at most ONE relationship
- result depends on which is Y
- less powerful in small samples

Johansen

a system (VECM-based) test

Best for

- THREE or more series
- finding several equilibria

How it works

- fits a VECM directly
- tests the rank of Π (trace / max-eig)

Strengths

- treats all series symmetrically
- finds the NUMBER of relations (rank)
- gives every cointegrating vector

Which Test, at a Glance

A one-look decision aid

Feature	Engle-Granger	Johansen
Number of series	2 (a pair)	2 or more
Approach	single-equation, 2-step	system (VECM-based)
Relations it finds	at most one	up to $k - 1$ (the rank)
Symmetric in variables	no (you pick Y)	yes

Use Engle-Granger for a fast pair check; use Johansen for many series or when the count of equilibria matters.

Practical Testing Pitfalls

Where cointegration tests mislead

Traps to watch

- Structural breaks can hide or fake cointegration.
- Short samples give low power — real links go undetected.
- The lag length in the test affects the result.

Good habits

- Confirm each series is $I(1)$ first, and inspect for breaks.
- Use enough data; be skeptical of borderline results.
- Choose lags by an information criterion, and check robustness.

The Cointegration Testing Workflow

Five steps to a defensible verdict

1 · Integration

each $I(1)$?

Unit-root test
each series;
confirm all are
 $I(1)$.

2 · Choose test

pair or system

Engle-Granger
for a pair;
Johansen for
more.

3 · Run it

proper values

Use the correct
(EG/Johansen)
critical values.

4 · Read result

rank / reject

Cointegrated?
How many
relations?

5 · Robustness

check

Vary lags,
orderings, sub-
samples; look
for breaks.

A positive result gives you β and the spread — the raw material for error correction and trading.

Test a Pair for Cointegration

Direct Claude Code through both tests, end to end

GOAL Run Engle-Granger and Johansen on a candidate pair, compare their verdicts, and read the rank.

[INPUT] Prompt · Claude Code

Write and run `coint_test.py` that:

1. confirms each series is $I(1)$ with ADF;
2. runs Engle-Granger BOTH ways ($Y \sim X$ and $X \sim Y$) using the proper critical values, reporting the p-values and estimated beta;
3. runs the Johansen test and prints the trace and max-eigenvalue statistics with the rank;
4. states the final verdict and the spread.

Note any disagreement between the two tests or between the two Engle-Granger orderings.

How to work with it

- Both series must be $I(1)$ for the test to be valid.
- EG and Johansen should broadly agree for a pair.
- The Johansen rank should be 0 or 1 for two series.
- Keep it; Part 3 trades the resulting spread.

Deliverable: `coint_test.py` + EG and Johansen results with a clear verdict.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	What exactly does Engle-Granger's Step 2 test?	Residual stationarity
Q2	Johansen returns rank $r = 2$ for four series. What does that mean?	Two equilibria
Q3	Why can't you use ordinary ADF critical values in Step 2?	Residual is estimated

All three come from the two-step logic, the rank, and the estimated-residual subtlety.

Three Misconceptions to Avoid

Where testing goes wrong

Common mistakes

- "A stationary-looking spread proves cointegration."
- "Plain ADF critical values are fine for Step 2."
- "Engle-Granger and Johansen are interchangeable."

The reality

- Appearances deceive over short samples — run the formal test.
- Use Engle-Granger critical values; the residual is estimated.
- EG suits a pair; Johansen suits systems and counts the rank.

Three Things to Carry Forward

Part 2 in one screen

1

Engle-Granger is a two-step residual test

Regress one level on the other to get β , then test whether the residual is stationary — with special critical values, because the residual is estimated.

2

Johansen tests a whole system

It fits a VECM and reads off the cointegration rank — how many independent equilibria exist — treating all series symmetrically. Best for three or more.

3

Prove it before you trade it

Confirm each series is $I(1)$, pick the right test, use the right critical values, and check robustness. A positive result hands you β and the spread.

Next — Part 3: error correction, the VECM, the speed of adjustment, and pairs trading Stage II.

★ WEEK 7 · PART 2 · PRACTICE

Testing for Cointegration, by Hand

The Engle-Granger Two-Step, and the Johansen Rank

From a residual test to a system verdict

Part 2's tests, with the arithmetic shown. We run the Engle-Granger two-step - regress, then test the residual against its special critical value - and then use the Johansen trace test to step down and count how many equilibria hold a pair together.

2P

PART 2 · APPLICATIONS

Prove It Before You Trade It

Case A: the Engle-Granger two-step - regress one price on the other, then test the residual for stationarity against the special critical value. Case B: the Johansen trace test, stepping down to find the cointegration rank. Then a Claude Code lab.

What Part 2 Told Us

Now we test it - with numbers

1

First confirm each series is $I(1)$

Cointegration is a statement about non-stationary series. Both prices must be $I(1)$ before any cointegration test makes sense.

2

Engle-Granger tests the residual

Regress one price on the other, then test the residual for stationarity - against special critical values, because the residual is estimated.

3

Johansen counts the equilibria

A symmetric system test finds the cointegration rank - how many independent stationary combinations hold the series together.

Everything here is a calculator exercise: one residual test, one trace-statistic step-down.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from two $I(1)$ price series.
- Run the residual test, or the trace test.
- Then read the verdict.

What you should watch

- Why the residual needs a special critical value.
- How the trace test steps down through ranks.
- How both reach the same conclusion.

The Engle-Granger Two-Step

Regress, then test the residual

Engle-Granger is the quick check for a single pair. Step one regresses one price on the other and keeps the residual - the spread. Step two runs an ADF test on that residual. Because the residual was estimated, we compare its statistic to a special, more demanding critical value. We work both steps and reach a verdict

The setup

- Two $I(1)$ prices, Y and X.
- 180 observations.
- A candidate pairs trade.

What we'll do by hand

- Regress Y on X for the spread.
- Run ADF on the residual.
- Compare to the Engle-Granger cutoff.

The Two-Step Procedure

A regression, then a residual test

Step 1: $Y_t = \alpha + \beta X_t + \hat{s}_t$ $\longrightarrow$ Step 2: ADF on $\hat{s}_t$ ($\Delta \hat{s}_t = \rho \hat{s}_{t-1} + \varepsilon_t$)

Step one

- Regress the level of Y on the level of X.
- The residual is the estimated spread.
- Beta is the cointegrating vector.

Step two

- Test that residual for a unit root.
- A stationary residual means cointegration.
- Use the special critical value.

Regress, Then Test

The spread and its ADF statistic

$$\text{Step 1: } Y_t = 3.65 + \mathbf{1.98} X_t + \hat{s}_t$$

$$\text{Step 2: ADF } t\text{-stat on } \hat{s}_t = \mathbf{-5.07}$$

The regression gives a hedge ratio of 1.98 and a residual spread. Running an ADF test on that spread yields a t-statistic of -5.07 - strongly negative, hinting the spread reverts. But we must check it against the right cutoff.

The Verdict

Against the special critical value

$$-5.07 < \underbrace{-3.34}_{\text{Engle-Granger 5\%}} < \underbrace{-2.86}_{\text{plain DF 5\%}} \Rightarrow \text{reject unit root} \Rightarrow \text{cointegrated}$$

Why a special value

- The residual was estimated, not observed.
- That makes stationarity easier to find by chance.
- So the cutoff is more negative than plain DF.

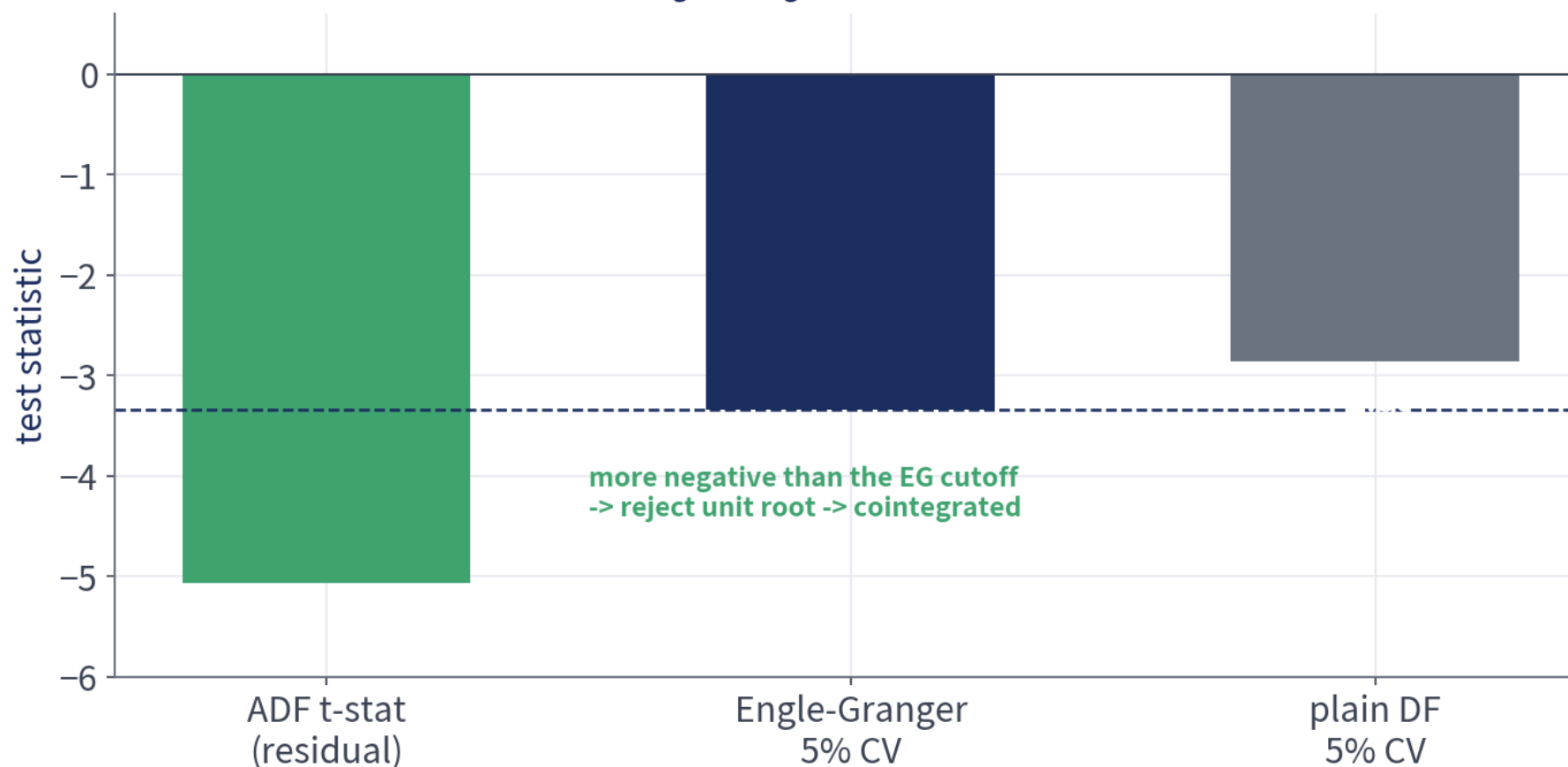
The decision

- At -5.07, the statistic clears -3.34.
- Reject the unit root - the spread is stationary.
- The pair is cointegrated.

Clearing the Right Bar

The residual's ADF statistic of -5.07 sits well below both cutoffs. Note that the Engle-Granger critical value (-3.34) is more demanding than plain Dickey-Fuller (-2.86) - using the wrong, easier bar would over-declare cointegration. Here the statistic clears even the tougher bar

Case A: the residual's ADF stat clears the Engle-Granger critical value



Simple, but Watch the Details

Two subtleties that matter

The special critical value

- Never use plain DF cutoffs on a residual.
- The estimated residual biases the test.
- Engle-Granger tables fix this.

The asymmetry

- Regressing X on Y gives a different beta (0.50).
- And a slightly different statistic (-5.04).
- Near the boundary, the two can disagree.

The Johansen Rank

Count the equilibria, symmetrically

Engle-Granger picks a dependent variable, so it isn't symmetric. Johansen treats all series alike and tests the whole system at once, returning the cointegration rank - the number of independent stationary combinations. We use the trace statistic and step down: reject rank zero, then stop when we can no longer reject

The rank r

- $r = 0$: no cointegration.
- $r = 1$: one equilibrium - a tradable pair.
- $r = 2$: both series already stationary.

What we'll do by hand

- Compute the trace statistic at $r = 0$.
- Then at r at most 1.
- Stop where we cannot reject.

The Trace Statistic

Step down until you cannot reject

$$\text{trace}(r) = -T \sum_{i=r+1}^k \ln(1 - \hat{\lambda}_i) \quad (\text{test } r=0, \text{ then } r \leq 1, \text{ step down until you cannot reject})$$

The procedure

- Start at rank zero - the null of no equilibrium.
- If the trace exceeds its critical value, reject.
- Move up a rank and test again.

The stopping rule

- Keep going until you cannot reject.
- That rank is the answer.
- It counts the cointegrating vectors.

Test Rank Zero

Is there any cointegration at all?

null hypothesis	trace statistic	5% critical value	decision
$r = 0$ (none)	28.6	15.49	REJECT

At rank zero, the trace statistic of 28.6 far exceeds the critical value of 15.49, so we reject the null of no cointegration. There is at least one equilibrium holding the pair together - now we ask whether there might be a second.

Trace 28.6 beats the 15.49 bar: reject no-cointegration, at least one equilibrium exists.

Step Up to Rank One

Where the step-down stops

null hypothesis	trace statistic	5% critical value	decision
$r \leq 1$ (one)	1.4	3.84	cannot reject

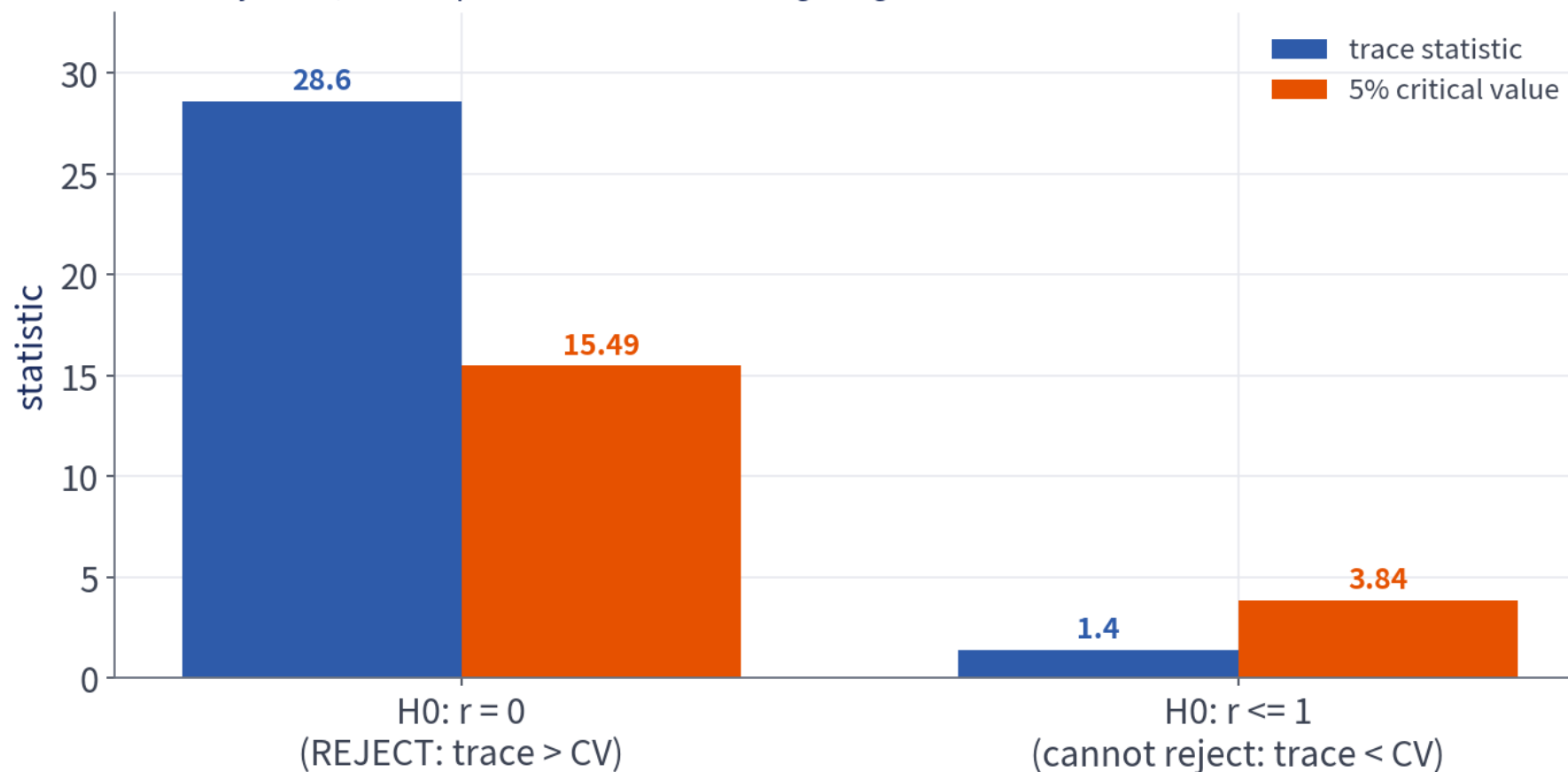
Now the trace statistic of 1.4 sits below the critical value of 3.84, so we cannot reject rank at most one. The step-down stops here: the rank is exactly one - a single cointegrating vector holds the pair together.

Fail to reject $r \leq 1$: the rank is exactly 1, so the pair is cointegrated.

Step Down to the Rank

At rank zero the trace statistic towers over its critical value, so we reject - there is cointegration. At rank at most one, the statistic drops below the bar, so we stop. The step-down lands on rank one: exactly one equilibrium - the symmetric confirmation of Case A's verdict

Case B: reject $r=0$, then stop at $r \leq 1 \rightarrow$ rank = 1 cointegrating vector



Why Reach for Johansen

When the system matters

Its advantages

- Symmetric - no dependent variable to pick.
- Handles many series at once.
- Counts multiple equilibria via the rank.

When it earns its keep

- Baskets of three or more assets.
- When Engle-Granger's asymmetry bites.
- When you need the number of equilibria.

Two Tests, One Verdict

Quick check on system test

Both routes ask the same question - is there a stationary combination? - and here both say yes. Engle-Granger is the fast two-step for a single pair, testing an estimated residual against a special cutoff. Johansen is the symmetric system test that counts equilibria via the rank. Use Engle-Granger to screen a pair, Johansen to

Engle-Granger (Case A)

- Regress, then ADF the residual.
- Statistic -5.07 clears -3.34.
- Fast, but asymmetric.

a

Johansen (Case B)

- Trace test, step down the rank.
- Reject $r=0$, stop at $r \leq 1$.
- Symmetric, counts equilibria.

Reproduce Both Tests

Direct Claude Code to check your arithmetic

GOAL Run the Engle-Granger two-step and the Johansen trace test on the same pair.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that, on a cointegrated pair Y, X (build one, or reuse Part 1's):

CASE A - Engle-Granger:

- * regress Y on X , keep the residual;
- * run `adfuller` on the residual (`regression='n'`);
- * print the t-stat and compare to -3.34 .

CASE B - Johansen:

- * run `coint_johansen(det_order=0, k_ar_diff=1)`;
- * print the trace stats and 5% critical values;
- * report the rank.

Confirm both say cointegrated.

Check against the slides

- $\beta = 1.98$, ADF -5.07 .
- clears EG -3.34 .
- trace $r=0$: $28.6 > 15.49$.
- trace $r \leq 1$: $1.4 < 3.84 \rightarrow r=1$.

Deliverable: `part2_practice.py` whose printed numbers match the two-step and the trace test.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why does Engle-Granger need special critical values?

The residual is estimated, biasing the test

Q2 How does Johansen determine the rank?

Step down until you cannot reject

Q3 When would you prefer Johansen?

For systems of three or more, or by symmetry

All three come from the Engle-Granger two-step and the Johansen trace test.

Three Pitfalls to Avoid

Where cointegration testing trips people

Common mistakes

- "Use plain Dickey-Fuller cutoffs on the residual."
- "Y-on-X and X-on-Y must give the same answer."
- "Skip checking that each series is $I(1)$."

The reality

- Use Engle-Granger critical values - the residual is fitted.
- The regression is asymmetric; results can differ.
- Cointegration only makes sense for $I(1)$ series.

Two Tests to Remember

Part 2 applications in one screen

1

Engle-Granger: regress, then test the residual

Regressing Y on X gave a hedge ratio of 1.98; the residual's ADF statistic of -5.07 cleared the special Engle-Granger cutoff of -3.34 - more demanding than plain DF because the residual is estimated. The pair is cointegrated.

2

Johansen: step down to the rank

The trace statistic of 28.6 rejected rank zero (critical 15.49), then 1.4 could not reject rank at most one (critical 3.84). The step-down stopped at rank one - a single cointegrating vector, symmetrically confirming the same verdict.

3

Screen with one, analyze with the other

Engle-Granger is a fast two-step for a single pair; Johansen is the symmetric system test that counts equilibria. Both agreed here - the pair is genuinely cointegrated and ready for the error-correction model.

Next — Part 3 practice: the error-correction model and building the pairs trade itself.

★ WEEK 7 · PART 3 of 3

Error Correction and Pairs Trading

The Restoring Force, and How to Trade It

The VECM · speed of adjustment · half-life · pairs trading Stage II

Cointegration says the spread reverts; error correction explains WHY. Part 3 turns that restoring force into a working strategy — and confronts the risk that the equilibrium can break.

3

PART 3 · ERROR CORRECTION

The Restoring Force, Traded

Why cointegrated series snap back, the vector error-correction model and its speed of adjustment, the half-life of reversion, and pairs trading built on the spread.

From an Equilibrium to a Mechanism

We proved the tie; now we model the pull

1

We can confirm cointegration

Parts 1 and 2 gave the concept and the tests. We know when a stationary spread — a long-run equilibrium — genuinely exists.

2

But WHAT keeps the spread from wandering?

If the pair deviates, something must pull it back. That something is error correction: an explicit restoring force in the dynamics.

3

And that force is exactly the trade

The speed and reliability of that pull decide whether a spread is tradable. Pairs trading Stage II is built directly on it.

Part 3 in one line: error correction is the restoring force — and the engine of pairs trading.

Error Correction: The Restoring Force

Why cointegrated series snap back

When a cointegrated pair drifts from its equilibrium, the next move tends to close the gap. That built-in correction is what keeps the spread stationary — a self-correcting system.

The mechanism

- A gap opens: the spread moves away from zero.
- The correction responds: subsequent changes shrink the gap.
- The bigger the gap, the stronger the pull back.

Why it must exist

- A stationary spread cannot wander off — so a force must restrain it.
- That force is the error-correction term.
- No such term, no cointegration — the two are inseparable.

Why the Gap Closes

The equilibrium error drives the next move

The feedback

- The system 'remembers' last period's deviation from equilibrium.
- It uses that memory to steer the next change.
- A positive gap prompts a downward correction, and vice versa.

What makes it a system

- Each series may adjust to help close the gap.
- The adjustments are proportional to the size of the deviation.
- Together they hold the pair near its long-run relationship.

The Vector Error-Correction Model

A VAR in differences, plus a memory of equilibrium

$$\Delta \mathbf{Y}_t = \alpha \underbrace{(\boldsymbol{\beta}' \mathbf{Y}_{t-1})}_{z_{t-1} \text{ (last gap)}} + \sum_i \Gamma_i \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t \quad \text{VECM}$$

The two parts

- The differenced terms are an ordinary VAR on the changes.
- The extra term is the LAST equilibrium error, $z(t-1)$.
- That term is what a plain VAR-in-differences leaves out.

Why it is the right model

- Differencing alone discards the long-run relationship.
- The VECM restores it as an explicit correction term.
- It is the natural model for cointegrated series.

Reading the VECM

One term does all the long-run work

The error-correction term

- It is last period's spread — the deviation from equilibrium.
- Its coefficient decides how strongly the series reacts.
- Everything else is just short-run VAR dynamics.

What the rest handles

- The differenced lags capture short-run momentum and noise.
- They matter for fit but not for the long-run tie.
- The long-run behavior lives entirely in the correction term.

The Speed of Adjustment

How fast the gap closes

$\alpha < 0$: restoring $|\alpha|$ large = fast correction, $|\alpha|$ small = slow

Reading alpha

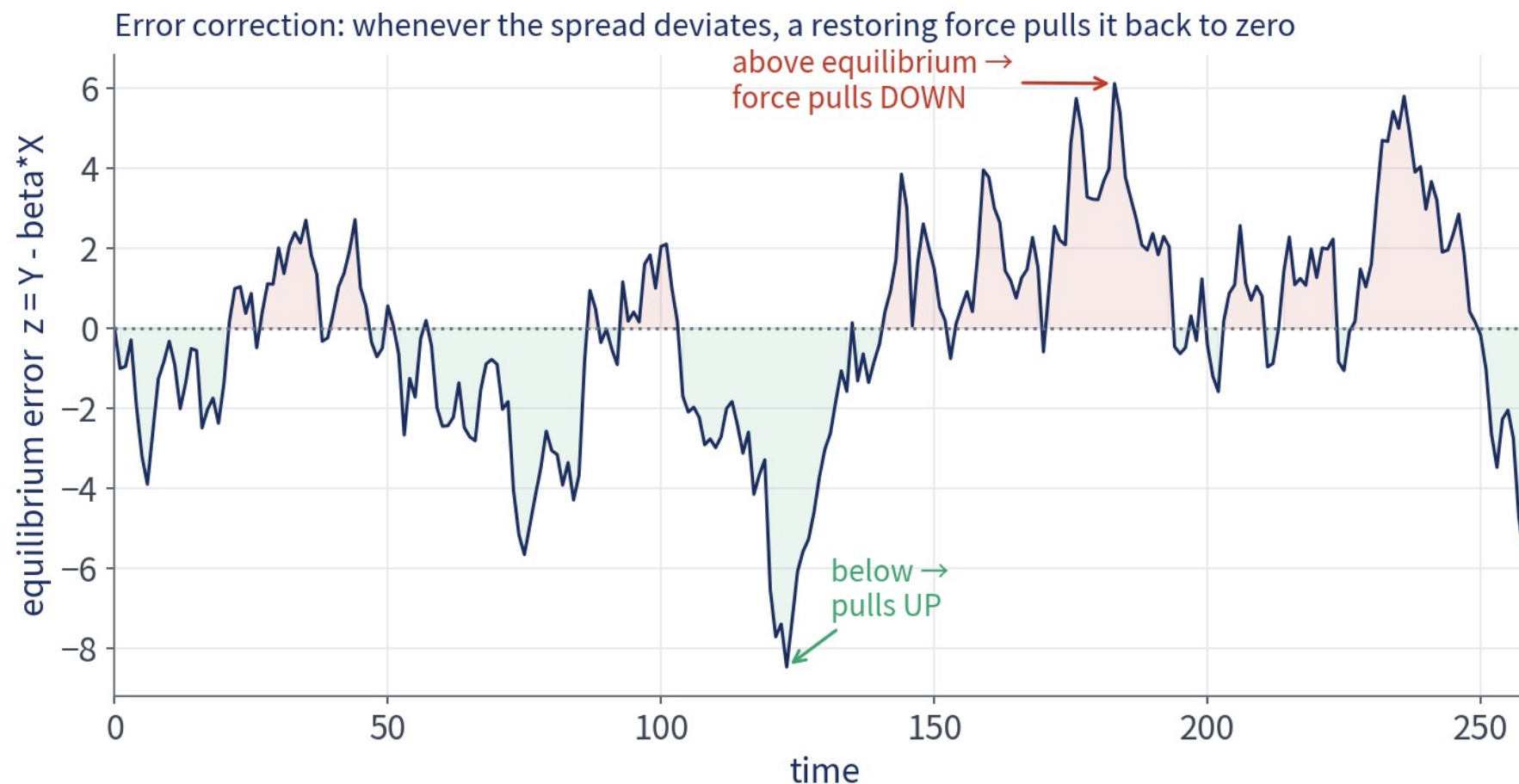
- It multiplies the last equilibrium error in the VECM.
- It must be negative to restore — pushing the gap toward zero.
- Its magnitude sets how much of the gap closes each period.

What it tells you

- A large $|\alpha|$ means fast, reliable reversion — attractive to trade.
- A small $|\alpha|$ means slow reversion — capital tied up longer.
- Which series has the bigger $|\alpha|$ shows who does the adjusting.

The Error-Correction Mechanism

Whenever the equilibrium error strays from zero, the restoring force acts: above the line it is pulled down, below it is pushed up. The gap is never allowed to run away.



Interpreting the Adjustment

Sign, size, and who moves

Sign and size

- Negative alpha restores; positive would be explosive (and rejected).
- Bigger $|\alpha|$ = faster correction, shorter round-trips.
- Near zero = a sluggish, barely-reverting spread.

Who adjusts

- Each series has its own alpha in the system.
- A series with alpha near zero barely moves to close the gap.
- The one with the larger $|\alpha|$ does most of the correcting.

Half-Life of Mean Reversion

How long until half the gap is gone?

$$\text{half-life} = \frac{\ln(0.5)}{\ln(1+\alpha)} \quad (\text{days for half the gap to close})$$

What it is

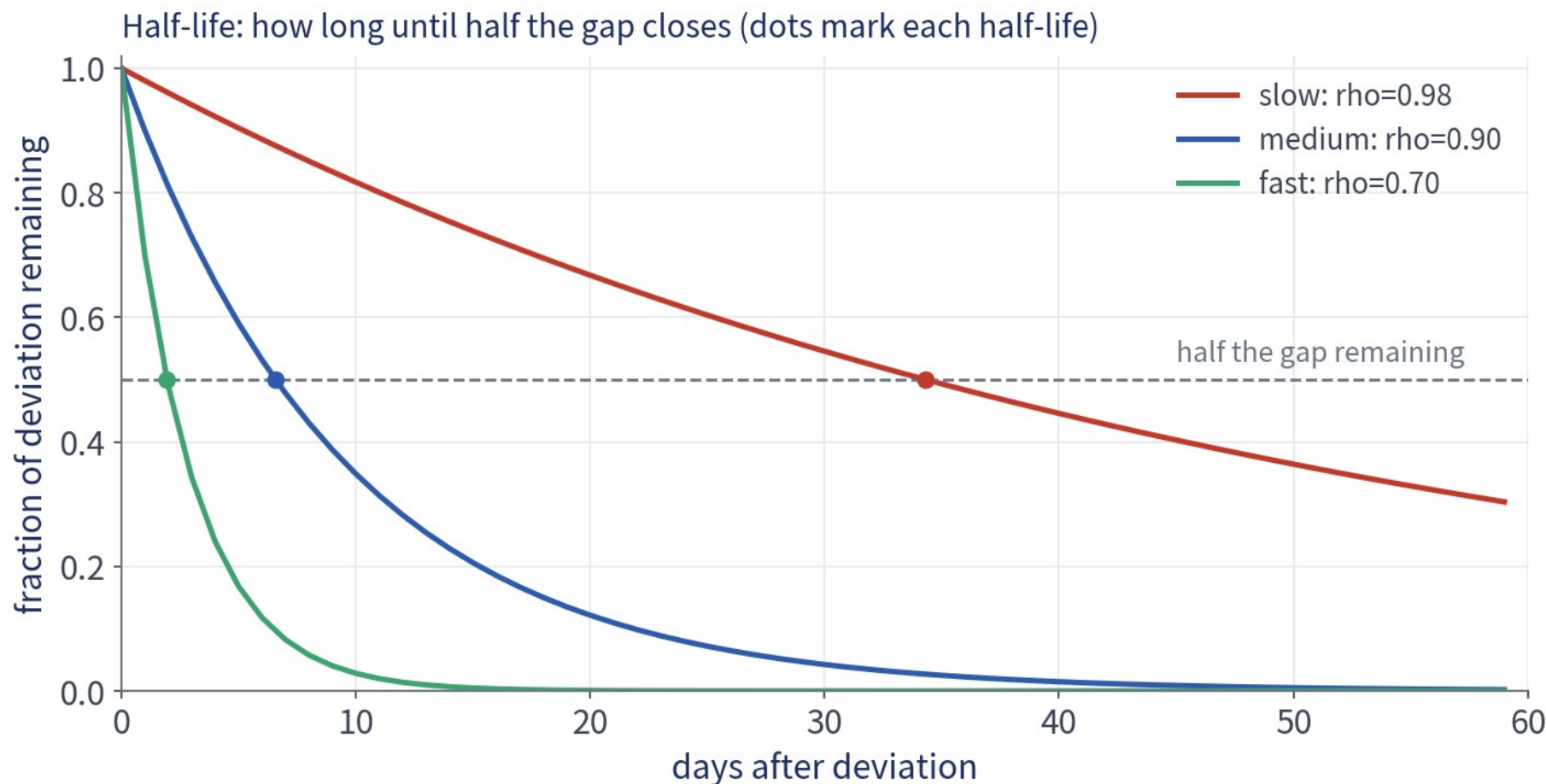
- The time for a deviation to shrink to half its size.
- It converts the abstract alpha into a plain horizon.
- A short half-life means quick, frequent opportunities.

Why traders use it

- It sets a sensible holding period for the trade.
- It flags spreads too slow to be worth the capital.
- It sanity-checks the model: is the speed plausible?

Fast vs Slow Reversion

A deviation decays geometrically; the dots mark each spread's half-life. Faster reversion (green) frees capital quickly; a very slow spread (red) may not be worth trading.



The Granger Representation Theorem

Cointegration and error correction are two sides of one coin

A landmark result: a set of series is cointegrated IF AND ONLY IF it can be written as an error-correction model. The equilibrium and the restoring force always come together.

What it guarantees

- Cointegration $\rightarrow$ an error-correction representation exists.
- An error-correction model $\rightarrow$ the series are cointegrated.
- You never have one without the other.

Why it matters

- It justifies modeling cointegrated series with a VECM.
- It links the long-run equilibrium to short-run dynamics.
- It is the theoretical backbone of this entire week.

Stage II: The Cointegration Spread

The thread gains its long-run anchor

Week 5 gave the pairs thread volatility-scaled bands; now cointegration gives it a genuine anchor. The spread is grounded in an equilibrium, not just a fitted correlation.

What Stage II adds

- A statistically validated, mean-reverting spread.
- A hedge ratio β from the cointegrating relationship.
- A restoring force with a measurable speed and half-life.

The trade in one sentence

- When the spread is stretched, fade it; as it reverts, take profit.
- Long the cheap leg, short the rich leg — market-neutral.
- Exit as the equilibrium error returns to zero.

The Z-Score Signal

Normalize the spread, then act on extremes

$$z_t = \frac{\text{spread}_t - \mu}{\sigma} \Rightarrow \text{enter at } |z_t| \geq 2, \text{ exit near } z_t = 0$$

How it works

- Standardize the spread into a z-score (distance from the mean).
- A large $|z|$ means the pair is unusually stretched.
- Enter against the stretch; exit as z returns to zero.

Why standardize

- It puts every pair on a common, comparable scale.
- Thresholds like 2 sigma become universal entry rules.
- It ties the signal directly to the equilibrium error.

Entry and Exit Rules

Turning the z-score into trades

Entry

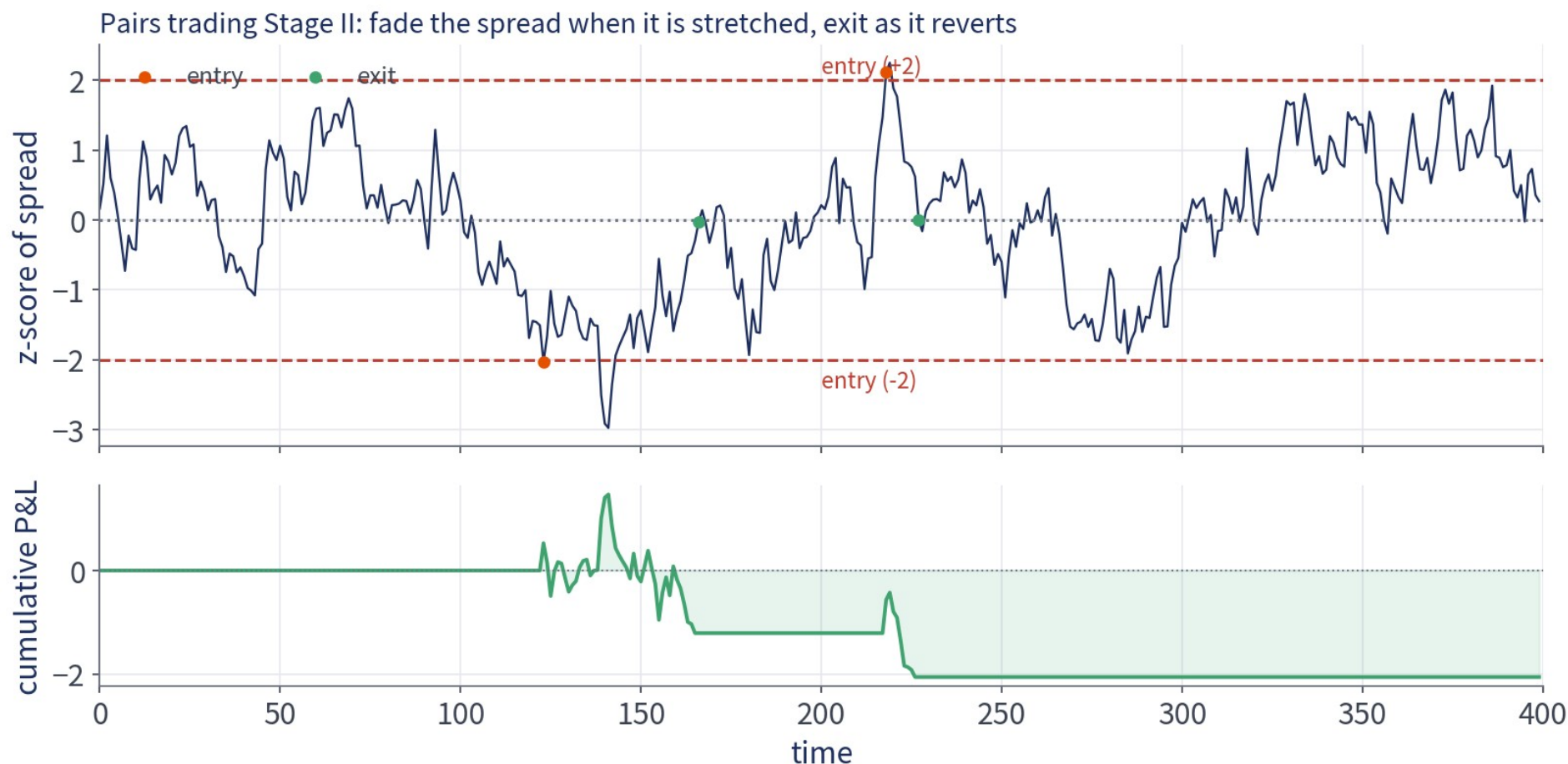
- Enter when the z-score exceeds a threshold (e.g. $|z| \geq 2$).
- Short the rich leg, long the cheap leg.
- Larger thresholds mean fewer but stronger signals.

Exit

- Exit as the spread reverts toward zero (the equilibrium).
- Some exit at $z = 0$; some take partial profits earlier.
- Always pair this with a stop for when reversion fails.

Pairs Trading in Action

Fade the spread when its z-score is stretched past the bands, exit as it reverts to zero. When the equilibrium holds, the cumulative P&L (below) grinds upward.



Sizing and Market Neutrality

The hedge ratio does the balancing

The hedge ratio

- The cointegrating β says how many units of X hedge one of Y.
- Trading in that ratio makes the position market-neutral.
- The bet is on the SPREAD, not on either market's direction.

Sizing the position

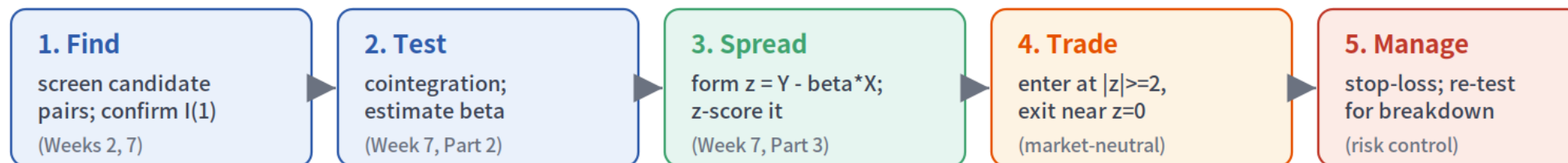
- Scale by the spread's volatility (Week 5) to target a risk level.
- Wider, riskier spreads take smaller notional.
- This is where volatility and cointegration finally combine.

The Complete Pairs-Trading Pipeline

Find a pair, test cointegration, form and z-score the spread, trade the extremes market-neutrally, and manage the risk — with error correction as the engine underneath.

The Complete Pairs-Trading Pipeline

Every week of the course feeds one line - from finding a pair to managing the trade.



The engine underneath

Cointegration gives the anchor (a stationary spread); error correction is WHY it reverts; volatility (Week 5) sizes the bands.

The ever-present risk

Cointegration can break. The spread may stop reverting and drift away - so re-test continuously and always keep a stop.

When Cointegration Breaks Down

The equilibrium is not guaranteed forever

A relationship that held for years can break — a merger, a regulatory change, a shifted business model. When it does, the spread stops reverting and drifts away, turning a 'sure' trade into a loss.

Why it happens

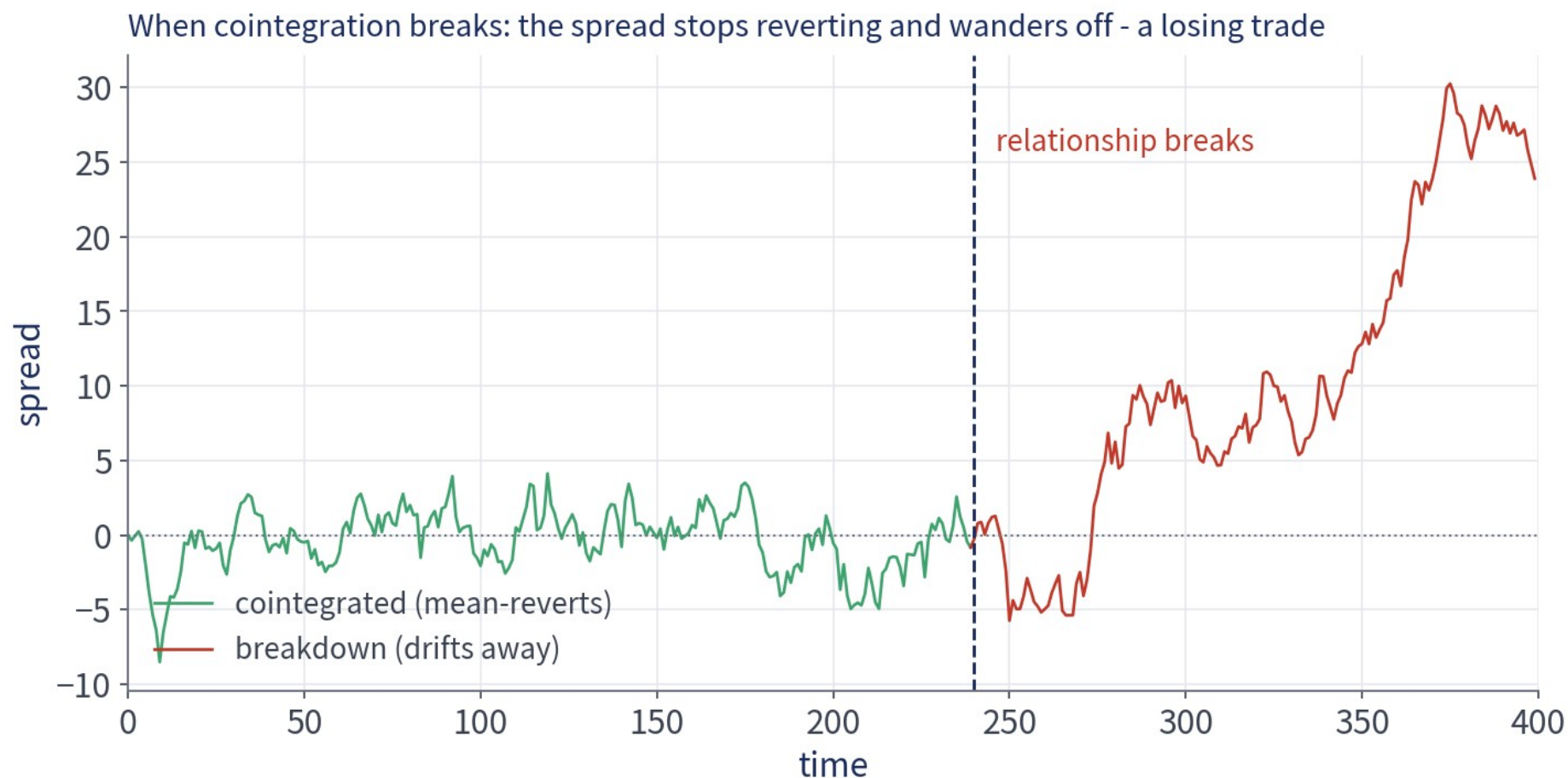
- The economic link that tied the pair changes or disappears.
- Structural breaks, regime shifts, or corporate events.
- The historical β is no longer the right relationship.

The danger

- The spread widens instead of reverting — losses compound.
- 'It always came back before' is a trap.
- This is the defining risk of statistical arbitrage.

When the Equilibrium Breaks

For a while the spread mean-reverts (green). Then the link breaks and it drifts away (red) — a widening, non-reverting spread that no longer offers a trade, only a loss.



Managing the Breakdown Risk

How to trade a spread you don't fully trust

Defenses

- Always run a stop-loss — cap the loss if reversion fails.
- Re-test cointegration regularly; exit if it weakens.
- Diversify across many pairs so one break isn't fatal.

Mindset

- Treat cointegration as a working assumption, not a law.
- Watch for the economic events that could break the link.
- Size positions so a broken pair is survivable.

Build the Spread and Trade It

Direct Claude Code through Pairs Trading Stage II

GOAL From a cointegrated pair, fit a VECM, extract the speed of adjustment and half-life, then back-test a z-score strategy.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage2.py` that:

1. takes a cointegrated pair, forms the spread $z = Y - \beta X$, and fits a VECM;
2. reports the speed of adjustment (α) and the implied half-life of mean reversion;
3. z-scores the spread and back-tests: enter at $|z| \geq 2$, exit near $z = 0$, market-neutral;
4. plots the z-score with signals and the cumulative P&L, and adds a stop-loss.

Report the half-life, the number of trades, and the P&L with vs without the stop.

How to work with it

- α should be negative (restoring).
- A short half-life means more, quicker trades.
- The stop should limit damage if the pair breaks.
- This completes the pairs-trading thread.

Deliverable: `pairs_stage2.py` + a VECM fit and a back-tested spread strategy with a stop.

Three Questions Before You Finish

If you can answer these, Week 7 has landed

Q1 In a VECM, what does a negative alpha coefficient represent?

**Restoring /
adjustment speed**

Q2 A spread's half-life is 3 days vs 60 days. Which is more tradable?

**3 days — faster
reversion**

Q3 Your pair's spread stops reverting and drifts. What has happened?

Cointegration broke

All three come from the speed of adjustment, the half-life, and the breakdown risk.

Three Pitfalls to Avoid

Where pairs trading goes wrong

Common mistakes

- "A cointegrated pair will always revert."
- "No stop-loss needed — the spread must come back."
- "The historical hedge ratio is fixed forever."

The reality

- Cointegration can break; re-test and never assume permanence.
- Always keep a stop — a broken pair drifts without limit.
- The relationship (and β) can drift; monitor and re-estimate.

Week 7 in Three Ideas

Pulling the whole week together

1

Cointegration is a stationary spread of wanderers

Part 1: two $I(1)$ series can share one common trend, leaving a stationary combination — the long-run equilibrium that cures spurious regression.

2

Two tests confirm it; the VECM explains it

Part 2: Engle-Granger and Johansen prove the equilibrium. Part 3: error correction is the restoring force, with a measurable speed and half-life.

3

The spread is the trade — but it can break

Pairs trading Stage II fades the z-scored spread market-neutrally. Its defining risk is that cointegration can break, so always re-test and keep a stop.

Next — Week 8: State-Space Models and the Kalman filter, plus the midterm.

★ WEEK 7 · PART 3 · PRACTICE

Error Correction and Pairs, by Hand

The Speed of Reversion, and the Trade It Powers

From the adjustment coefficient to a z-score trade

Part 3's payoff, with the arithmetic shown. We read the error-correction coefficient to find how fast a spread reverts and its half-life, then turn the spread into a z-score signal - entering, exiting, and stopping - to run the pairs trade the whole week built toward.

3P

PART 3 · APPLICATIONS

From Restoring Force to Trade

Case A: the speed of adjustment - how fast the gap closes - and the half-life of mean reversion. Case B: normalizing the spread into a z-score and turning it into entry, exit, and stop rules with a worked profit. Then a Claude Code lab.

What Part 3 Told Us

Now we trade it - with numbers

1

Error correction is the restoring force

When cointegrated series drift apart, the equilibrium error pulls them back. The adjustment coefficient measures how hard.

2

The half-life sets the clock

From the adjustment speed, the half-life tells you how many days until half the gap is gone - your expected holding period.

3

The z-score turns the spread into a trade

Normalize the spread, fade the extremes, and exit at the mean. Entry, exit, and stop rules follow directly.

Everything here is a calculator exercise: one adjustment speed, one half-life, one round-trip trade.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the spread and its dynamics.
- Read the adjustment speed, or the z-score.
- Then act on the number.

What you should watch

- How a negative alpha closes the gap.
- How the half-life sets the holding period.
- How the z-score triggers the trade.

The Speed of Reversion

How fast the gap closes

When the spread strays from equilibrium, error correction pulls it back. The adjustment coefficient - alpha, negative - is the fraction of the gap erased each day. From it we get the half-life: how long until half the deviation is gone. That number is the expected holding period of a pairs trade, so it's worth computing exactly

The setup

- The spread from our cointegrated pair.
- Its error-correction coefficient is -0.25.
- It reverts toward equilibrium each day.

What we'll do by hand

- Read alpha as a fraction closed per day.
- Convert to the AR(1) coefficient.
- Compute the half-life.

The Error-Correction Model

A restoring force toward equilibrium

$$\Delta Y_t = \underbrace{\alpha (Y_{t-1} - \beta X_{t-1})}_{\text{pull back to equilibrium}} + (\text{short-run terms}) + \varepsilon_t \quad (\alpha < 0 \text{ is the restoring force})$$

The mechanism

- Last period's equilibrium error drives today.
- Alpha scales how strongly it pulls back.
- A negative alpha means a restoring force.

Who adjusts

- Each series has its own alpha.
- The larger one does more of the correcting.
- Sometimes only one side moves.

Read the Adjustment Speed

A fraction of the gap, per day

$$\alpha = -0.25 \Rightarrow \text{25\% of the gap closes each day,} \quad \phi = 1 + \alpha = 0.75 \text{ (spread AR(1))}$$

The sign and size

- Negative alpha pulls the spread back.
- Its magnitude is 0.25.
- So a quarter of the gap closes each day.

The AR(1) view

- The spread follows an AR(1) with $\phi = 0.75$.
- Each day it keeps 75% of its deviation.
- That sets up the half-life.

The Half-Life

How long until half the gap is gone

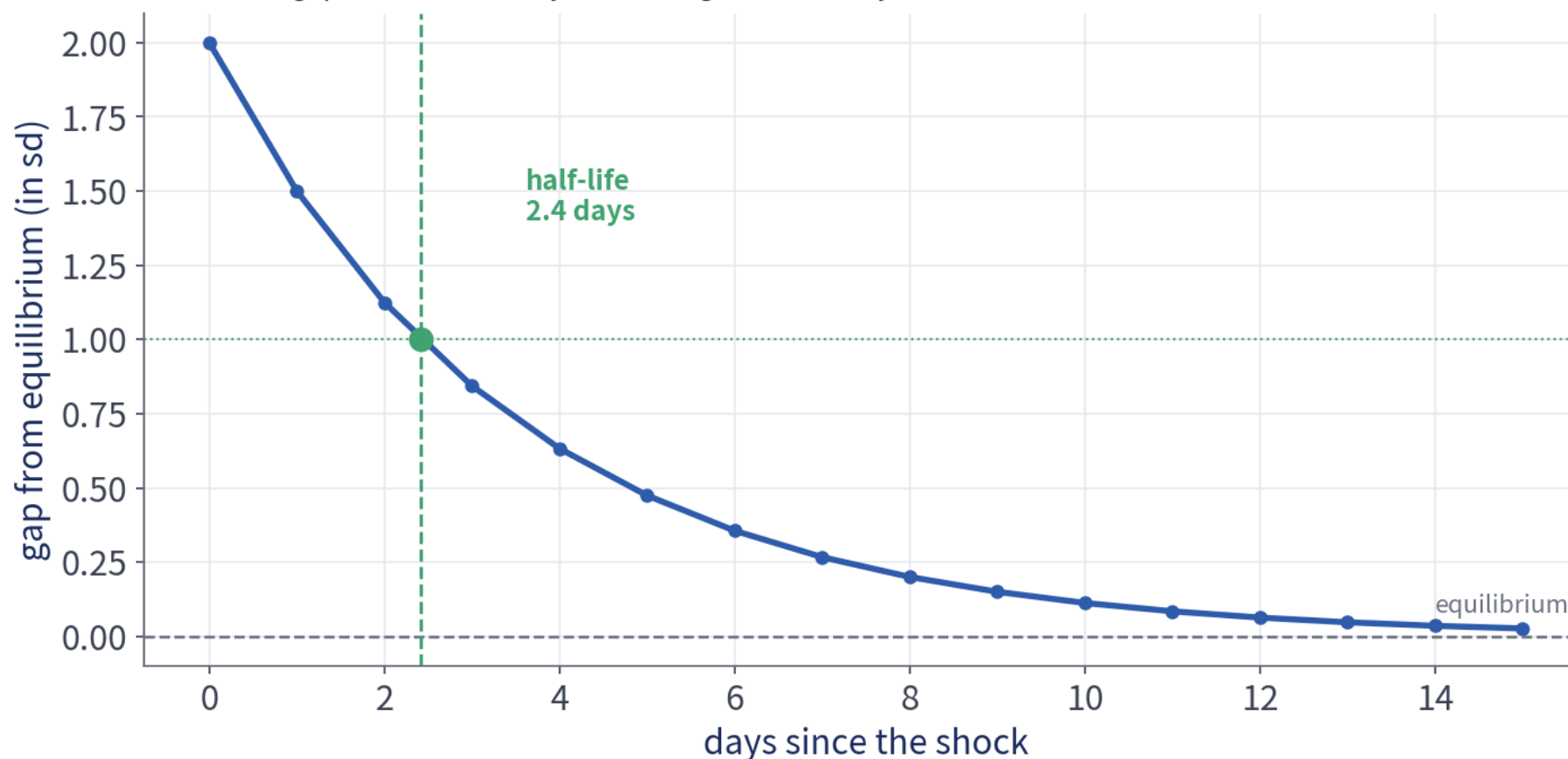
$$\text{half-life} = \frac{\ln 0.5}{\ln \phi} = \frac{\ln 0.5}{\ln 0.75} = \frac{-0.693}{-0.288} = \mathbf{2.4} \text{ days}$$

Dividing the log of one-half by the log of the AR(1) coefficient gives a half-life of 2.4 days. Half of any deviation is erased in under three days - a fast-reverting spread, ideal for a pairs trade.

The Gap Decays Away

Starting two standard deviations from equilibrium, the spread shrinks 25% a day, tracing a smooth exponential decay back to zero. It crosses the halfway line at 2.4 days - the half-life. The faster this decay, the shorter you hold each trade and the more often you can trade

Case A: the gap closes 25% a day - half of it gone in 2.4 days



The Half-Life Is the Clock

It sets the whole trade

What it tells you

- The expected holding period per trade.
- How many round trips a year you can make.
- Whether the edge is worth the capital.

What to avoid

- A near-zero alpha means glacial reversion.
- Long half-lives tie up capital for little.
- Prefer pairs that snap back quickly.

The Z-Score Trade

Normalize the spread, then act

With the hedge ratio and a fast half-life in hand, the trade is simple.

Standardize the spread into a z-score, so its units are standard deviations. When the z-score reaches plus or minus two, the spread is stretched - fade it. When it returns to zero, exit. If it pushes past three, the equilibrium may be breaking - cut the loss. We work a full

The spread stats

- Mean 5.2, standard deviation 0.62.
- The z-score rescales it in sigmas.
- Extremes are trade signals.

What we'll do by hand

- Find the entry and exit spread levels.
- Compute the captured spread.
- Set the stop.

The Z-Score Signal

Standard deviations from the mean

$$z_t = \frac{s_t - \bar{s}}{\sigma_s} = \frac{s_t - 5.2}{0.62} : \quad z = +2 \leftrightarrow s = 6.44, \quad z = 0 \leftrightarrow s = 5.20$$

Reading the z-score

- Zero means the spread is at equilibrium.
- Plus or minus two means it's stretched two sigma.
- The sign says which way to trade.

Turning it into levels

- A z of +2 is a spread of 6.44.
- A z of 0 is a spread of 5.20.
- Those are the entry and exit prices.

Entry, Exit, and Stop

Turning z into a trade plan

z-score	spread level	action
+2	6.44	ENTER: short 1 Y, long 1.96 X
0	5.20	EXIT: spread back at the mean
+3	7.06	STOP: equilibrium may be breaking

At a z of +2 the spread is stretched to 6.44 - short it. Exit when it reverts to 5.20 at z of zero. If instead it widens to 7.06 at z of +3, the pair may be decoupling: stop out.

The Round-Trip Profit

What the reversion pays

enter at $z = +2$ ($s=6.44$) : short 1 Y , long 1.96 X

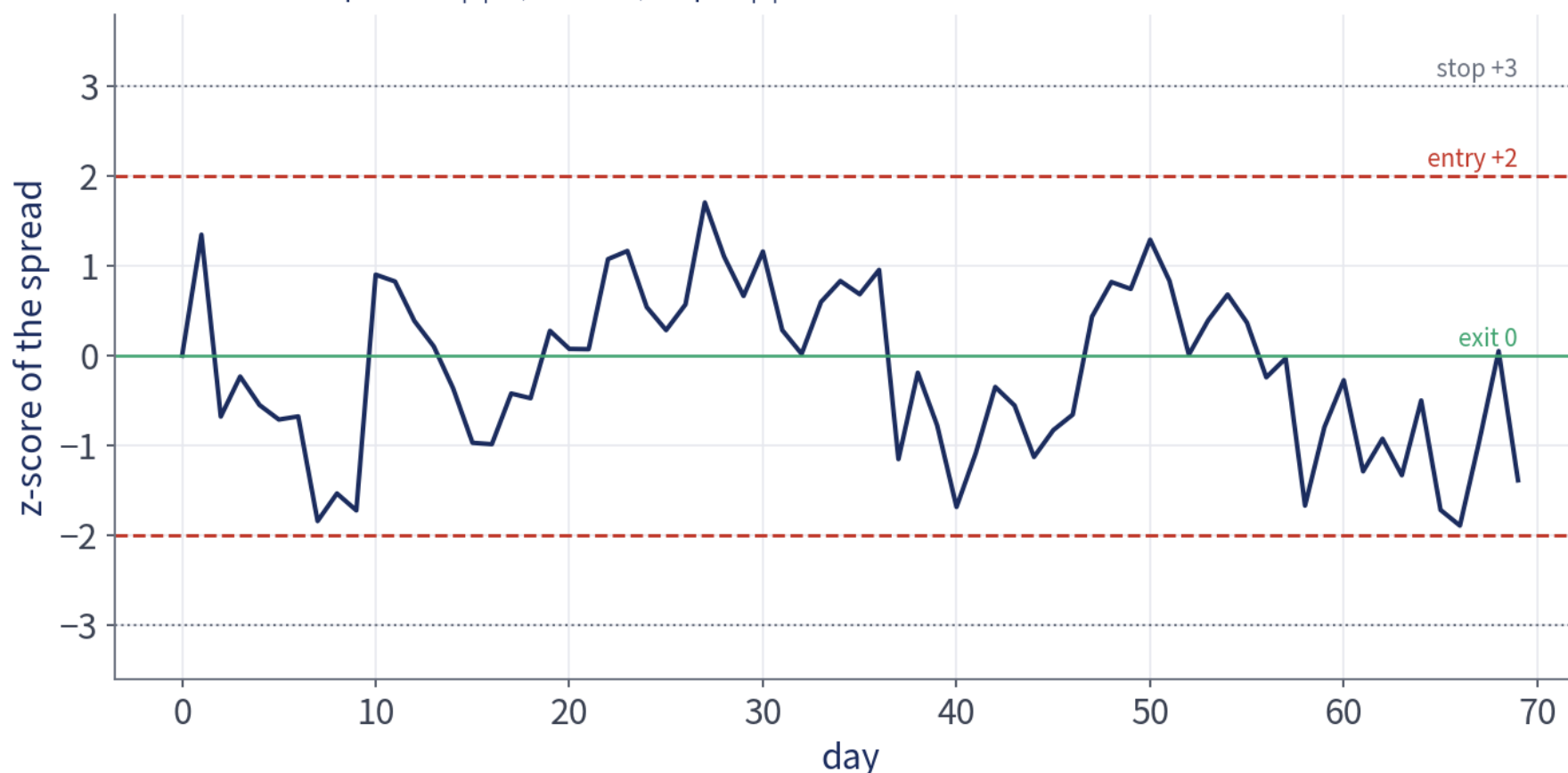
exit at $z=0$ ($s=5.20$) : captured $6.44 - 5.20 = \mathbf{1.24}$ per share of Y

Shorting the spread at 6.44 and covering at 5.20 captures 1.24 points per share of Y . With a 2.4-day half-life, the position reverts in about five days - a fast, repeatable round trip.

Fade the Extremes, Exit at Zero

The z-score oscillates around zero. Each time it pierces the plus-or-minus-two bands (red), the spread is stretched - enter against it. Exit as it returns to the green zero line. The dotted plus-or-minus-three lines are the stop: beyond them, the equilibrium may be breaking down

Case B: fade the spread at $|z|=2$, exit at 0, stop at $|z|=3$



The Trade, and Its Guardrail

Profit rests on the spread reverting

Why it works

- Cointegration guarantees the spread reverts.
- The z-score times entry and exit cleanly.
- The half-life bounds the holding period.

The guardrail

- Cointegration can break - watch for it.
- A z past three is a warning, not an opportunity.
- The stop caps the loss when it does.

The Full Pairs Recipe

Everything Week 7 Unit 1

The whole week comes together here. Confirm the pair is cointegrated, estimate the hedge ratio, and form the spread. Read the adjustment speed for the half-life. Standardize into a z-score, then fade the extremes, exit at the mean, and stop if the equilibrium breaks. Cointegration is the foundation; error

The reversion (Case A)

- Alpha of -0.25 closes 25% a day.
- Half-life of 2.4 days.
- A fast, tradable spread.

The trade (Case B)

- Enter at z of +2, spread 6.44.
- Exit at z of 0, capturing 1.24.
- Stop at z of +3.

; t

Reproduce the Reversion and the Trade

Direct Claude Code to check your arithmetic

GOAL Compute the half-life, then backtest the z-score rules on a cointegrated spread.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that, on a cointegrated spread (reuse Part 1/2's, mean 5.2, sd 0.62):

CASE A - reversion:

- * fit the spread's AR(1); print alpha, phi, and the half-life $\ln(0.5)/\ln(\phi)$.

CASE B - the trade:

- * compute the z-score;
- * enter at $|z|=2$, exit at 0, stop at $|z|=3$;
- * print the entry/exit spread levels and the captured points per round trip.

Then plot the decay and the z-score with the bands.

Check against the slides

- alpha -0.25, phi 0.75.
- half-life 2.4 days.
- entry 6.44, exit 5.20.
- captured 1.24 per trade.

Deliverable: `part3_practice.py` whose printed numbers match the half-life and the round-trip trade.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a negative adjustment coefficient mean?

**A restoring force
pulling the spread
back**

Q2 How do you get the half-life from the AR(1) coefficient?

**Log of one-half over
log of phi**

Q3 When do you stop out of a pairs trade?

**When the z-score
keeps widening past
the stop**

All three come from the adjustment coefficient, the half-life, and the z-score rules.

Three Pitfalls to Avoid

Where pairs trading trips people

Common mistakes

- "A stretched spread must revert soon."
- "Hold through any widening - it always comes back."
- "The hedge ratio is fixed forever."

The reality

- A long half-life means it may take weeks.
- Cointegration can break; the stop is essential.
- Re-estimate beta as the relationship drifts.

Two Calculations to Remember

Part 3 applications in one screen

1

The adjustment speed sets the half-life

An error-correction coefficient of -0.25 closes 25% of the gap each day, an AR(1) coefficient of 0.75. The half-life - $\log$ of one-half over $\log$ of 0.75 - is 2.4 days, the expected holding period of a trade in this spread.

2

The z-score turns the spread into a trade

Standardizing the spread (mean 5.2, sd 0.62), a z of +2 is a spread of 6.44 - short it - and a z of 0 is 5.20, where you cover, capturing 1.24 points. A z of +3 at 7.06 is the stop, in case the equilibrium breaks.

3

Cointegration, error correction, the z-score

The pair's shared trend is the foundation, error correction is the engine that reverts the spread, and the z-score is the trigger that times entry, exit, and the stop. That is the full pairs recipe.

Week 7 complete — from the idea of cointegration, through testing it, to the error-correction model and a live pairs trade.